

PYTHON-01

PART 5

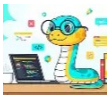
EXCEPTIONS, FILES, DATA, TIME, LOGGING, RELIABILITY & SECURE OPERATION

Chapters 150-262 | 113 Chapters | EN / FR

Engineering Reference • AHDS



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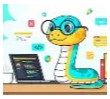
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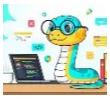
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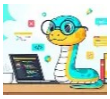
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Chapter 150 - Errors vs Exceptions

ENGLISH	FRANÇAIS
What does Errors vs Exceptions actually do? A syntax error prevents compilation, while a runtime exception is an object raised during execution; distinguish programmer defects from recoverable environmental faults.	**Quel est le rôle de Errors vs Exceptions ?** Une erreur de syntaxe empêche l'analyse du code; une exception apparaît à l'exécution. Distinguer défaut de programmation et incident externe récupérable.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

ENGINEERING NOTE / NOTE TECHNIQUE

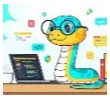
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CODE / API EXAMPLE

```
try:
    value = int("bad")
except ValueError as exc:
    print(type(exc).__name__)
```



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Chapter 151 - Exception Hierarchy

ENGLISH	FRANÇAIS
What does Exception Hierarchy actually do? Most application exceptions inherit Exception; BaseException also includes control signals such as KeyboardInterrupt and SystemExit, which should not normally be swallowed.	**Quel est le rôle de Exception Hierarchy ?** La plupart des exceptions applicatives héritent de Exception; BaseException inclut aussi KeyboardInterrupt et SystemExit, à ne pas intercepter sans raison.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

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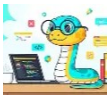
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CODE / API EXAMPLE

```
try:
   int("bad")
except Exception as exc:
   print(type(exc).__mro__[:3])
```



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Chapter 152 - try / except

ENGLISH	FRANÇAIS
What does try / except actually do? A try block limits the operation being guarded; an except block handles only the failures it understands, preserving all others for higher layers.	**Quel est le rôle de try / except ?** Le bloc try délimite l'opération risquée; except traite seulement les erreurs prévues et laisse remonter les autres.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

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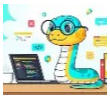
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CODE / API EXAMPLE

```
try:
    number = int("12")
except ValueError:
    number = None
print(number)
```



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Chapter 153 - Multiple except Clauses

ENGLISH	FRANÇAIS
What does Multiple except Clauses actually do? Separate except clauses let the program respond differently to conversion errors, absent files and permission failures; order handlers from specific to broad.	**Quel est le rôle de Multiple except Clauses ?** Des clauses except distinctes associent une réponse adaptée à chaque panne; placer les cas précis avant les cas généraux.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

ENGINEERING NOTE / NOTE TECHNIQUE

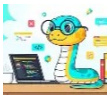
ENGINEERING NOTE / NOTE TECHNIQUE: Separate except clauses let the program respond differently to conversion errors, absent files and permission failures; order handlers from specific to broad. / Des clauses except distinctes associent une réponse adaptée à chaque panne; placer les cas précis avant les cas généraux.

CODE / API EXAMPLE

```
try:
    with open("missing.csv", encoding="utf-8") as f:
        text = f.read()
except FileNotFoundError:
    text = "missing"
except PermissionError:
    text = "denied"
```



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Chapter 154 - Handling Multiple Exception Types

ENGLISH	FRANÇAIS
What does Handling Multiple Exception Types actually do? A tuple in except catches several exception classes with one policy; use it only when those faults genuinely require the same recovery action.	**Quel est le rôle de Handling Multiple Exception Types ?** Un tuple dans except regroupe plusieurs types d'exception lorsque leur traitement est réellement identique.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

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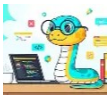
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CODE / API EXAMPLE

```
try:
    value = int("bad")
except (ValueError, TypeError) as exc:
    print("invalid input:", type(exc).__name__)
```



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Chapter 155 - else in Exception Handling

ENGLISH	FRANÇAIS
What does else in Exception Handling actually do? An else suite runs only when the try suite completed without an exception; put post-processing there rather than expanding the protected region.	**Quel est le rôle de else in Exception Handling ?** La clause else s'exécute uniquement si try réussit; y placer le traitement suivant pour éviter d'intercepter ses propres erreurs.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

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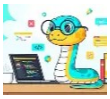
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CODE / API EXAMPLE

```
try:
    result = int("42")
except ValueError:
    print("invalid")
else:
    print("accepted", result)
```



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Chapter 156 - finally

ENGLISH	FRANÇAIS
What does finally actually do? finally normally runs whether execution succeeds or fails, making it useful for releasing resources; it is not a substitute for context managers and does not survive abrupt process termination.	**Quel est le rôle de finally ?** finally s'exécute normalement avec ou sans exception pour libérer les ressources; il ne protège pas contre l'arrêt brutal du processus.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

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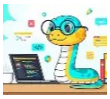
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CODE / API EXAMPLE

```
resource = [1]
try:
    print(resource[0])
finally:
    resource.clear()
```



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Chapter 157 - Nested Exception Handling

ENGLISH	FRANÇAIS
What does Nested Exception Handling actually do? Nested handlers support layered recovery: a device adapter may translate an I/O error while a supervisory layer decides whether to retry or stop.	**Quel est le rôle de Nested Exception Handling ?** Les gestionnaires imbriqués séparent traduction d'erreur et décision de reprise, sans masquer l'exception d'origine.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

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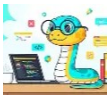
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CODE / API EXAMPLE

```
try:
   try:
      raise OSError("device offline")
   except OSError as exc:
      raise RuntimeError("acquisition failed") from exc
except RuntimeError as exc:
   print(exc.__cause__)
```



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Chapter 158 - raise

ENGLISH	FRANÇAIS
What does raise actually do? raise creates or propagates an exception when a contract fails; select a meaningful built-in or custom exception with an actionable message.	**Quel est le rôle de raise ?** raise signale explicitement un contrat non respecté; choisir une classe précise et un message exploitable.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

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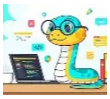
ENGINEERING NOTE / NOTE TECHNIQUE: raise creates or propagates an exception when a contract fails; select a meaningful built-in or custom exception with an actionable message. / raise signale explicitement un contrat non respecté; choisir une classe précise et un message exploitable.

CODE / API EXAMPLE

```
def validate_voltage(v):
   if not 0 <= v <= 3.3:
      raise ValueError("voltage outside ADC range")
validate_voltage(2.5)
```



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Chapter 159 - Re-Raising Exceptions

ENGLISH	FRANÇAIS
What does Re-Raising Exceptions actually do? A bare raise inside an active exception handler rethrows the same exception with its traceback; writing raise exc can alter traceback presentation.	**Quel est le rôle de Re-Raising Exceptions ?** Un raise sans argument dans except relance la même exception et conserve son contexte de pile.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

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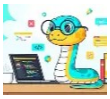
ENGINEERING NOTE / NOTE TECHNIQUE: A bare raise inside an active exception handler rethrows the same exception with its traceback; writing raise exc can alter traceback presentation. / Un raise sans argument dans except relance la même exception et conserve son contexte de pile.

CODE / API EXAMPLE

```
try:
   try:
      int("bad")
   except ValueError:
      raise
except ValueError:
   print("propagated")
```



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Chapter 160 - Exception Chaining

ENGLISH	FRANÇAIS
What does Exception Chaining actually do? Exception chaining preserves the low-level cause while presenting a domain-specific error; use raise NewError(...) from exc at API boundaries.	**Quel est le rôle de Exception Chaining ?** Le chaînage conserve la cause technique tout en présentant une erreur métier; employer raise ... from exc.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

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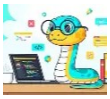
ENGINEERING NOTE / NOTE TECHNIQUE: Exception chaining preserves the low-level cause while presenting a domain-specific error; use raise NewError(...) from exc at API boundaries. / Le chaînage conserve la cause technique tout en présentant une erreur métier; employer raise ... from exc.

CODE / API EXAMPLE

```
try:
    int("bad")
except ValueError as exc:
    try:
        raise RuntimeError("invalid configuration") from exc
    except RuntimeError as wrapped:
        print(type(wrapped.__cause__).__name__)
```



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Chapter 161 - Exception Context & Explicit Causes

ENGLISH	FRANÇAIS
What does Exception Context & Explicit Causes actually do? __context__ records implicit handling context; __cause__ is the explicit cause set by from, and 'from None' suppresses its display, not necessarily its storage.	**Quel est le rôle de Exception Context & Explicit Causes ?** __context__ est implicite, __cause__ explicite via from; from None masque l'affichage du contexte sans toujours l'effacer.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

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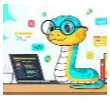
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CODE / API EXAMPLE

```
try:
   int("no")
except ValueError as first:
   try:
      raise RuntimeError("bad calibration") from first
   except RuntimeError as second:
      print(second.__cause__ is first)
```



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Chapter 162 - Custom Exceptions

ENGLISH	FRANÇAIS
What does Custom Exceptions actually do? A custom exception makes a stable domain-level failure contract; inherit from Exception and avoid encoding all state only in a human-readable message.	**Quel est le rôle de Custom Exceptions ?** Une exception personnalisée fournit un contrat d'erreur stable; hériter de Exception et conserver les données structurées utiles.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

ENGINEERING NOTE / NOTE TECHNIQUE

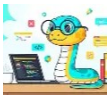
ENGINEERING NOTE / NOTE TECHNIQUE: A custom exception makes a stable domain-level failure contract; inherit from Exception and avoid encoding all state only in a human-readable message. / Une exception personnalisée fournit un contrat d'erreur stable; hériter de Exception et conserver les données structurées utiles.

CODE / API EXAMPLE

```
class CalibrationError(Exception):
    pass
try:
    raise CalibrationError("reference expired")
except CalibrationError as exc:
    print(exc)
```



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Chapter 163 - Custom Exception Hierarchies

ENGLISH	FRANÇAIS
What does Custom Exception Hierarchies actually do? Create a narrow exception hierarchy so callers can catch a common device-error base or a specific timeout without confusing unrelated faults.	**Quel est le rôle de Custom Exception Hierarchies ?** Une hiérarchie limitée permet d'intercepter une famille de pannes ou un cas précis sans confondre les origines.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

ENGINEERING NOTE / NOTE TECHNIQUE

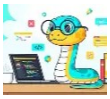
ENGINEERING NOTE / NOTE TECHNIQUE: Create a narrow exception hierarchy so callers can catch a common device-error base or a specific timeout without confusing unrelated faults. / Une hiérarchie limitée permet d'intercepter une famille de pannes ou un cas précis sans confondre les origines.

CODE / API EXAMPLE

```
class DeviceError(Exception): pass
class DeviceTimeout(DeviceError): pass
try:
    raise DeviceTimeout("CH2")
except DeviceError as exc:
    print(type(exc).__name__)
```



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Chapter 164 - Exception Objects

ENGLISH	FRANÇAIS
What does Exception Objects actually do? Exception instances carry args, type and optional attributes; attach structured metadata such as channel ID without placing sensitive values in logs.	**Quel est le rôle de Exception Objects ?** Les objets exception possèdent args et des attributs facultatifs; ajouter un identifiant de voie sans divulguer de secrets.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

ENGINEERING NOTE / NOTE TECHNIQUE

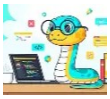
ENGINEERING NOTE / NOTE TECHNIQUE: Exception instances carry args, type and optional attributes; attach structured metadata such as channel ID without placing sensitive values in logs. / Les objets exception possèdent args et des attributs facultatifs; ajouter un identifiant de voie sans divulguer de secrets.

CODE / API EXAMPLE

```
class ChannelError(Exception):
    def __init__(self, channel):
        super().__init__(f"channel {channel} failed")
        self.channel = channel
print(ChannelError(3).channel)
```



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Chapter 165 - Tracebacks

ENGLISH	FRANÇAIS
What does Tracebacks actually do? A traceback identifies the call stack at failure; preserve it when logging and translate it carefully across library or thread boundaries.	**Quel est le rôle de Tracebacks ?** Un traceback localise la pile d'appels; le conserver pour le diagnostic tout en évitant les données sensibles.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

ENGINEERING NOTE / NOTE TECHNIQUE

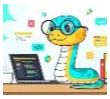
ENGINEERING NOTE / NOTE TECHNIQUE: A traceback identifies the call stack at failure; preserve it when logging and translate it carefully across library or thread boundaries. / Un traceback localise la pile d'appels; le conserver pour le diagnostic tout en évitant les données sensibles.

CODE / API EXAMPLE

```
import traceback
try:
    1 / 0
except ZeroDivisionError:
    print(traceback.format_exc(limit=1))
```



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Chapter 166 - ExceptionGroup

ENGLISH	FRANÇAIS
What does ExceptionGroup actually do? ExceptionGroup aggregates independent failures, as in concurrent acquisition; partial except* handling leaves unmatched subexceptions to propagate.	**Quel est le rôle de ExceptionGroup ?** ExceptionGroup regroupe des erreurs indépendantes; except* ne traite que les sous-groupes correspondants.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

ENGINEERING NOTE / NOTE TECHNIQUE

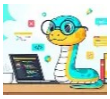
ENGINEERING NOTE / NOTE TECHNIQUE: ExceptionGroup aggregates independent failures, as in concurrent acquisition; partial except* handling leaves unmatched subexceptions to propagate. / ExceptionGroup regroupe des erreurs indépendantes; except* ne traite que les sous-groupes correspondants.

CODE / API EXAMPLE

```
try:
    raise ExceptionGroup("channels", [ValueError("A"), TimeoutError("B")])
except* ValueError as group:
    print("value errors:", len(group.exceptions))
except* TimeoutError as group:
    print("timeouts:", len(group.exceptions))
```



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Chapter 167 - except*

ENGLISH	FRANÇAIS
What does except* actually do? except* splits an exception group by matching type and can run more than one handler; it is not interchangeable with ordinary except in the same try suite.	**Quel est le rôle de except* ?** except* répartit les sous-exceptions par type et plusieurs clauses peuvent s'exécuter; ne pas le mélanger avec except dans le même try.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

ENGINEERING NOTE / NOTE TECHNIQUE

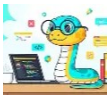
ENGINEERING NOTE / NOTE TECHNIQUE: except* splits an exception group by matching type and can run more than one handler; it is not interchangeable with ordinary except in the same try suite. / except* répartit les sous-exceptions par type et plusieurs clauses peuvent s'exécuter; ne pas le mélanger avec except dans le même try.

CODE / API EXAMPLE

```
try:
   raise ExceptionGroup("batch", [ValueError("bad"), OSError("io")])
except* ValueError:
   print("validation subset")
except* OSError:
   print("I/O subset")
```



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Chapter 168 - assert

ENGLISH	FRANÇAIS
What does assert actually do? assert checks developer assumptions and may be removed with Python -O; never rely on it to validate configuration, user input or safety limits.	**Quel est le rôle de assert ?** assert sert aux invariants de développement et peut disparaître avec -O; ne pas l'utiliser pour les contrôles de sécurité.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

ENGINEERING NOTE / NOTE TECHNIQUE

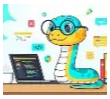
ENGINEERING NOTE / NOTE TECHNIQUE: assert checks developer assumptions and may be removed with Python -O; never rely on it to validate configuration, user input or safety limits. / assert sert aux invariants de développement et peut disparaître avec -O; ne pas l'utiliser pour les contrôles de sécurité.

CODE / API EXAMPLE

```
samples = [1.0, 2.0]
assert len(samples) == 2
print(sum(samples))
```



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Chapter 169 - Assertions vs Runtime Validation

ENGLISH	FRANÇAIS
What does Assertions vs Runtime Validation actually do? Runtime validation must execute regardless of optimization; use explicit if/raise for measurements, permissions and untrusted data.	**Quel est le rôle de Assertions vs Runtime Validation ?** La validation d'exécution doit rester active en mode optimisé; utiliser if et raise pour les entrées externes.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

ENGINEERING NOTE / NOTE TECHNIQUE

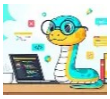
ENGINEERING NOTE / NOTE TECHNIQUE: Runtime validation must execute regardless of optimization; use explicit if/raise for measurements, permissions and untrusted data. / La validation d'exécution doit rester active en mode optimisé; utiliser if et raise pour les entrées externes.

CODE / API EXAMPLE

```
def sample_rate(rate):
    if not isinstance(rate, int) or rate <= 0:
        raise ValueError("rate must be positive integer")
    return rate
print(sample_rate(1000))
```



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Chapter 170 - Exception-Handling Strategies

ENGLISH	FRANÇAIS
What does Exception-Handling Strategies actually do? Define error taxonomy, observability and recovery per boundary: retry transient communication faults, reject invalid commands and fail safely on unknown states.	**Quel est le rôle de Exception-Handling Strategies ?** Définir une politique d'erreurs par frontière: réessayer une panne transitoire, refuser une commande invalide et arrêter sans danger si l'état est inconnu.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

ENGINEERING NOTE / NOTE TECHNIQUE

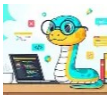
ENGINEERING NOTE / NOTE TECHNIQUE: Define error taxonomy, observability and recovery per boundary: retry transient communication faults, reject invalid commands and fail safely on unknown states. / Définir une politique d'erreurs par frontière: réessayer une panne transitoire, refuser une commande invalide et arrêter sans danger si l'état est inconnu.

CODE / API EXAMPLE

```
def parse_count(s):
   try: return int(s)
   except ValueError as exc:
      raise ValueError("invalid sample count") from exc
```



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Chapter 171 - Exceptions at Hardware & Communication Boundaries

ENGLISH	FRANÇAIS
What does Exceptions at Hardware & Communication Boundaries actually do? At serial/USB/network boundaries, distinguish timeout, disconnection and malformed frames; do not report stale data as a new valid measurement.	**Quel est le rôle de Exceptions at Hardware & Communication Boundaries ?** Aux interfaces série/USB/réseau, distinguer délai, déconnexion et trame invalide; ne jamais présenter une donnée périmée comme récente.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

ENGINEERING NOTE / NOTE TECHNIQUE

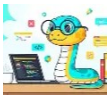
ENGINEERING NOTE / NOTE TECHNIQUE: At serial/USB/network boundaries, distinguish timeout, disconnection and malformed frames; do not report stale data as a new valid measurement. / Aux interfaces série/USB/réseau, distinguer délai, déconnexion et trame invalide; ne jamais présenter une donnée périmée comme récente.

CODE / API EXAMPLE

```
class LinkTimeout(Exception): pass
def read_frame(payload):
    if not payload: raise LinkTimeout("no frame")
    if payload[0] != 0xAA: raise ValueError("sync byte")
    return payload[1:]
print(read_frame(b"\xAA\x01"))
```



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Chapter 172 - Error Codes vs Exceptions

ENGLISH	FRANÇAIS
What does Error Codes vs Exceptions actually do? Return status codes are useful across C ABI/MCU interfaces, while exceptions suit Python control flow; translate codes once with a documented mapping.	**Quel est le rôle de Error Codes vs Exceptions ?** Les codes d'état conviennent aux ABI C et MCU; traduire les codes en exceptions Python à une frontière explicite.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

ENGINEERING NOTE / NOTE TECHNIQUE

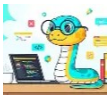
ENGINEERING NOTE / NOTE TECHNIQUE: Return status codes are useful across C ABI/MCU interfaces, while exceptions suit Python control flow; translate codes once with a documented mapping. / Les codes d'état conviennent aux ABI C et MCU; traduire les codes en exceptions Python à une frontière explicite.

CODE / API EXAMPLE

```
def check_status(code):
    if code == 0: return
    if code == 1: raise TimeoutError("device timeout")
    raise OSError(f"device status {code}")
check_status(0)
```



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Chapter 173 - Warnings & Warning Categories - warnings

ENGLISH	FRANÇAIS
What does Warnings & Warning Categories - warnings actually do? warnings communicate non-fatal concerns such as deprecated calibration; choose category and severity so critical invalid measurements become errors instead.	**Quel est le rôle de Warnings & Warning Categories - warnings ?** warnings signale des anomalies non fatales comme un étalonnage ancien; une mesure invalide exige plutôt une erreur.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

ENGINEERING NOTE / NOTE TECHNIQUE

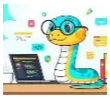
ENGINEERING NOTE / NOTE TECHNIQUE: warnings communicate non-fatal concerns such as deprecated calibration; choose category and severity so critical invalid measurements become errors instead. / warnings signale des anomalies non fatales comme un étalonnage ancien; une mesure invalide exige plutôt une erreur.

CODE / API EXAMPLE

```
import warnings
warnings.warn("calibration due", RuntimeWarning, stacklevel=1)
```



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Chapter 174 - Warning Filters, Suppression & Engineering Use

ENGLISH	FRANÇAIS
What does Warning Filters, Suppression & Engineering Use actually do? Use local catch_warnings or targeted filters; suppress only well-understood categories because global suppression may hide actionable defects.	**Quel est le rôle de Warning Filters, Suppression & Engineering Use ?** Utiliser catch_warnings et des filtres ciblés; une suppression globale peut cacher des défauts importants.
What is the engineering design constraint? Capture only understood failures; preserve diagnostics and propagate unknown errors.	**Quelle contrainte de conception faut-il respecter ?** Intercepter seulement les erreurs prévues, conserver le diagnostic et propager les autres.
How should the behavior be verified? Exercise a normal result, an expected exception, and an unexpected exception.	**Comment vérifier le comportement ?** Tester réussite, exception prévue et exception inattendue.

ENGINEERING NOTE / NOTE TECHNIQUE

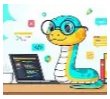
ENGINEERING NOTE / NOTE TECHNIQUE: Use local catch_warnings or targeted filters; suppress only well-understood categories because global suppression may hide actionable defects. / Utiliser catch_warnings et des filtres ciblés; une suppression globale peut cacher des défauts importants.

CODE / API EXAMPLE

```
import warnings
with warnings.catch_warnings(record=True) as captured:
    warnings.simplefilter("always", RuntimeWarning)
    warnings.warn("drift", RuntimeWarning)
print(len(captured))
```



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Chapter 175 - Resource Lifetime & Cleanup

ENGLISH	FRANÇAIS
What does Resource Lifetime & Cleanup actually do? Resource ownership includes files, locks, ports and native handles; close deterministically even if an operation raises.	**Quel est le rôle de Resource Lifetime & Cleanup ?** La propriété d'une ressource couvre fichiers, verrous, ports et handles; libérer même lorsqu'une opération échoue.
What is the engineering design constraint? Bind each resource to one explicit owner and release it on all exit paths.	**Quelle contrainte de conception faut-il respecter ?** Attribuer chaque ressource à un propriétaire et la libérer sur chaque sortie.
How should the behavior be verified? Inject an exception halfway through acquisition and verify deterministic cleanup.	**Comment vérifier le comportement ?** Injecter une panne et vérifier la libération de toutes les ressources.

ENGINEERING NOTE / NOTE TECHNIQUE

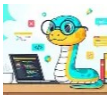
ENGINEERING NOTE / NOTE TECHNIQUE: Resource ownership includes files, locks, ports and native handles; close deterministically even if an operation raises. / La propriété d'une ressource couvre fichiers, verrous, ports et handles; libérer même lorsqu'une opération échoue.

CODE / API EXAMPLE

```
from contextlib import closing
from io import BytesIO
with closing(BytesIO(b"ADC")) as buf:
    print(buf.read())
```



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Chapter 176 - The with Statement

ENGLISH	FRANÇAIS
What does The with Statement actually do? with calls __enter__ and __exit__ and normally closes the resource on any exit; an __exit__ returning true can suppress an exception, so design carefully.	**Quel est le rôle de The with Statement ?** with appelle __enter__ et __exit__; un __exit__ qui renvoie vrai peut supprimer une exception, ce qui doit être intentionnel.
What is the engineering design constraint? Bind each resource to one explicit owner and release it on all exit paths.	**Quelle contrainte de conception faut-il respecter ?** Attribuer chaque ressource à un propriétaire et la libérer sur chaque sortie.
How should the behavior be verified? Inject an exception halfway through acquisition and verify deterministic cleanup.	**Comment vérifier le comportement ?** Injecter une panne et vérifier la libération de toutes les ressources.

ENGINEERING NOTE / NOTE TECHNIQUE

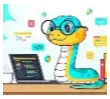
ENGINEERING NOTE / NOTE TECHNIQUE: with calls __enter__ and __exit__ and normally closes the resource on any exit; an __exit__ returning true can suppress an exception, so design carefully. / with appelle __enter__ et __exit__; un __exit__ qui renvoie vrai peut supprimer une exception, ce qui doit être intentionnel.

CODE / API EXAMPLE

```
from io import StringIO
with StringIO("measurement") as f:
    print(f.read())
```



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Chapter 177 - Multiple Context Managers

ENGLISH	FRANÇAIS
What does Multiple Context Managers actually do? Multiple context managers acquire left-to-right and release right-to-left; failure to acquire a later resource still releases earlier ones.	**Quel est le rôle de Multiple Context Managers ?** Les contextes multiples s'ouvrent de gauche à droite et se ferment dans l'ordre inverse, même si une acquisition échoue.
What is the engineering design constraint? Bind each resource to one explicit owner and release it on all exit paths.	**Quelle contrainte de conception faut-il respecter ?** Attribuer chaque ressource à un propriétaire et la libérer sur chaque sortie.
How should the behavior be verified? Inject an exception halfway through acquisition and verify deterministic cleanup.	**Comment vérifier le comportement ?** Injecter une panne et vérifier la libération de toutes les ressources.

ENGINEERING NOTE / NOTE TECHNIQUE

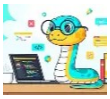
ENGINEERING NOTE / NOTE TECHNIQUE: Multiple context managers acquire left-to-right and release right-to-left; failure to acquire a later resource still releases earlier ones. / Les contextes multiples s'ouvrent de gauche à droite et se ferment dans l'ordre inverse, même si une acquisition échoue.

CODE / API EXAMPLE

```
from io import StringIO
with StringIO("A") as a, StringIO("B") as b:
    print(a.read(), b.read())
```



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Chapter 178 - Creating Context Managers

ENGLISH	FRANÇAIS
What does Creating Context Managers actually do? Define a context manager via __enter__/__exit__ or @contextmanager; guarantee cleanup and do not swallow unexpected exceptions.	**Quel est le rôle de Creating Context Managers ?** Créer un contexte avec __enter__/__exit__ ou @contextmanager; garantir la libération sans masquer les erreurs.
What is the engineering design constraint? Bind each resource to one explicit owner and release it on all exit paths.	**Quelle contrainte de conception faut-il respecter ?** Attribuer chaque ressource à un propriétaire et la libérer sur chaque sortie.
How should the behavior be verified? Inject an exception halfway through acquisition and verify deterministic cleanup.	**Comment vérifier le comportement ?** Injecter une panne et vérifier la libération de toutes les ressources.

ENGINEERING NOTE / NOTE TECHNIQUE

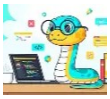
ENGINEERING NOTE / NOTE TECHNIQUE: Define a context manager via __enter__/__exit__ or @contextmanager; guarantee cleanup and do not swallow unexpected exceptions. / Créer un contexte avec __enter__/__exit__ ou @contextmanager; garantir la libération sans masquer les erreurs.

CODE / API EXAMPLE

```
from contextlib import contextmanager
@contextmanager
def session():
   print("open")
   try: yield "device"
   finally: print("close")
with session() as dev: print(dev)
```



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Chapter 179 - contextlib

ENGLISH	FRANÇAIS
What does contextlib actually do? contextlib offers ExitStack, closing, suppress and contextmanager; ExitStack is useful when a dynamic number of devices must be acquired safely.	**Quel est le rôle de contextlib ?** contextlib fournit ExitStack, closing, suppress et contextmanager; ExitStack gère un nombre variable de ressources.
What is the engineering design constraint? Bind each resource to one explicit owner and release it on all exit paths.	**Quelle contrainte de conception faut-il respecter ?** Attribuer chaque ressource à un propriétaire et la libérer sur chaque sortie.
How should the behavior be verified? Inject an exception halfway through acquisition and verify deterministic cleanup.	**Comment vérifier le comportement ?** Injecter une panne et vérifier la libération de toutes les ressources.

ENGINEERING NOTE / NOTE TECHNIQUE

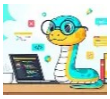
ENGINEERING NOTE / NOTE TECHNIQUE: contextlib offers ExitStack, closing, suppress and contextmanager; ExitStack is useful when a dynamic number of devices must be acquired safely. / contextlib fournit ExitStack, closing, suppress et contextmanager; ExitStack gère un nombre variable de ressources.

CODE / API EXAMPLE

```
from contextlib import ExitStack
from io import StringIO
with ExitStack() as stack:
    ports = [stack.enter_context(StringIO(str(i))) for i in range(2)]
    print([p.read() for p in ports])
```



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Chapter 180 - Context Managers for Hardware Resources

ENGLISH	FRANÇAIS
What does Context Managers for Hardware Resources actually do? A hardware resource context should configure a safe state, own the connection, and release outputs/locks on failure; actual driver APIs must define fail-safe behavior.	**Quel est le rôle de Context Managers for Hardware Resources ?** Un contexte matériel doit établir un état sûr, gérer la connexion et libérer sorties/verrous; le pilote réel doit définir le repli sûr.
What is the engineering design constraint? Bind each resource to one explicit owner and release it on all exit paths.	**Quelle contrainte de conception faut-il respecter ?** Attribuer chaque ressource à un propriétaire et la libérer sur chaque sortie.
How should the behavior be verified? Inject an exception halfway through acquisition and verify deterministic cleanup.	**Comment vérifier le comportement ?** Injecter une panne et vérifier la libération de toutes les ressources.

ENGINEERING NOTE / NOTE TECHNIQUE

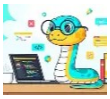
ENGINEERING NOTE / NOTE TECHNIQUE: A hardware resource context should configure a safe state, own the connection, and release outputs/locks on failure; actual driver APIs must define fail-safe behavior. / Un contexte matériel doit établir un état sûr, gérer la connexion et libérer sorties/verrous; le pilote réel doit définir le repli sûr.

CODE / API EXAMPLE

```
from contextlib import contextmanager
@contextmanager
def output():
   state = {"enabled": True}
   try: yield state
   finally: state["enabled"] = False
with output() as device: print(device["enabled"])
```



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Chapter 181 - File I/O Fundamentals

ENGLISH	FRANÇAIS
What does File I/O Fundamentals actually do? File I/O is a boundary where permissions, partial writes, encoding and filesystem failures affect data integrity; define the format and failure policy first.	**Quel est le rôle de File I/O Fundamentals ?** L'E/S fichier dépend des droits, écritures partielles, encodage et erreurs disque; définir format et politique de panne.
What is the engineering design constraint? Specify the external file contract and guard against truncation, platform differences and interruption.	**Quelle contrainte de conception faut-il respecter ?** Définir le contrat fichier et protéger contre troncature, différences OS et interruption.
How should the behavior be verified? Test missing, empty, truncated and read-only files as appropriate.	**Comment vérifier le comportement ?** Tester fichiers absents, vides, tronqués et accès refusé selon le cas.

ENGINEERING NOTE / NOTE TECHNIQUE

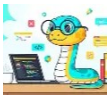
ENGINEERING NOTE / NOTE TECHNIQUE: File I/O is a boundary where permissions, partial writes, encoding and filesystem failures affect data integrity; define the format and failure policy first. / L'E/S fichier dépend des droits, écritures partielles, encodage et erreurs disque; définir format et politique de panne.

CODE / API EXAMPLE

```
from io import BytesIO
stream = BytesIO(b"DATA")
assert stream.read(4) == b"DATA"
```



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Chapter 182 - open()

ENGLISH	FRANÇAIS
What does open() actually do? open creates a text or binary stream; specify mode and encoding explicitly and close it reliably using with.	**Quel est le rôle de open() ?** open ouvre un flux texte ou binaire; préciser mode et encodage, puis fermer avec with.
What is the engineering design constraint? Specify the external file contract and guard against truncation, platform differences and interruption.	**Quelle contrainte de conception faut-il respecter ?** Définir le contrat fichier et protéger contre troncature, différences OS et interruption.
How should the behavior be verified? Test missing, empty, truncated and read-only files as appropriate.	**Comment vérifier le comportement ?** Tester fichiers absents, vides, tronqués et accès refusé selon le cas.

ENGINEERING NOTE / NOTE TECHNIQUE

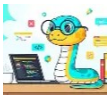
ENGINEERING NOTE / NOTE TECHNIQUE: open creates a text or binary stream; specify mode and encoding explicitly and close it reliably using with. / open ouvre un flux texte ou binaire; préciser mode et encodage, puis fermer avec with.

CODE / API EXAMPLE

```
from pathlib import Path
p = Path("example.txt")
with p.open("w", encoding="utf-8") as f: f.write("OK")
print(p.read_text(encoding="utf-8"))
```



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Chapter 183 - File Modes

ENGLISH	FRANÇAIS
What does File Modes actually do? File modes r/w/a/x control creation and truncation; b selects binary while + requests read-write, so select mode to prevent accidental data loss.	**Quel est le rôle de File Modes ?** Les modes r/w/a/x gèrent lecture, troncature, ajout et création exclusive; b impose le binaire et + lecture-écriture.
What is the engineering design constraint? Specify the external file contract and guard against truncation, platform differences and interruption.	**Quelle contrainte de conception faut-il respecter ?** Définir le contrat fichier et protéger contre troncature, différences OS et interruption.
How should the behavior be verified? Test missing, empty, truncated and read-only files as appropriate.	**Comment vérifier le comportement ?** Tester fichiers absents, vides, tronqués et accès refusé selon le cas.

ENGINEERING NOTE / NOTE TECHNIQUE

ENGINEERING NOTE / NOTE TECHNIQUE: File modes r/w/a/x control creation and truncation; b selects binary while + requests read-write, so select mode to prevent accidental data loss. / Les modes r/w/a/x gèrent lecture, troncature, ajout et création exclusive; b impose le binaire et + lecture-écriture.

CODE / API EXAMPLE

```
from tempfile import TemporaryFile
with TemporaryFile(mode="w+t", encoding="utf-8") as f:
    f.write("V=3.3\n"); f.seek(0)
    print(f.readline().strip())
```



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Chapter 184 - Text Files

ENGLISH	FRANÇAIS
What does Text Files actually do? Text I/O performs character encoding and newline translation; specify UTF-8 and error policy for interoperable engineering logs.	**Quel est le rôle de Text Files ?** L'E/S texte encode les caractères et adapte les fins de ligne; fixer UTF-8 et une stratégie d'erreurs.
What is the engineering design constraint? Specify the external file contract and guard against truncation, platform differences and interruption.	**Quelle contrainte de conception faut-il respecter ?** Définir le contrat fichier et protéger contre troncature, différences OS et interruption.
How should the behavior be verified? Test missing, empty, truncated and read-only files as appropriate.	**Comment vérifier le comportement ?** Tester fichiers absents, vides, tronqués et accès refusé selon le cas.

ENGINEERING NOTE / NOTE TECHNIQUE

ENGINEERING NOTE / NOTE TECHNIQUE: Text I/O performs character encoding and newline translation; specify UTF-8 and error policy for interoperable engineering logs. / L'E/S texte encode les caractères et adapte les fins de ligne; fixer UTF-8 et une stratégie d'erreurs.

CODE / API EXAMPLE

```
from io import StringIO
with StringIO() as f:
   f.write("température = 24 °C")
   print(f.getvalue())
```



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Chapter 185 - Binary Files

ENGLISH	FRANÇAIS
What does Binary Files actually do? Binary streams operate on bytes without implicit character conversion; use them for firmware images, raw sensor frames and checksums.	**Quel est le rôle de Binary Files ?** Les flux binaires manipulent des octets sans conversion; utiles pour firmware, trames capteur et sommes de contrôle.
What is the engineering design constraint? Specify the external file contract and guard against truncation, platform differences and interruption.	**Quelle contrainte de conception faut-il respecter ?** Définir le contrat fichier et protéger contre troncature, différences OS et interruption.
How should the behavior be verified? Test missing, empty, truncated and read-only files as appropriate.	**Comment vérifier le comportement ?** Tester fichiers absents, vides, tronqués et accès refusé selon le cas.

ENGINEERING NOTE / NOTE TECHNIQUE

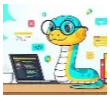
ENGINEERING NOTE / NOTE TECHNIQUE: Binary streams operate on bytes without implicit character conversion; use them for firmware images, raw sensor frames and checksums. / Les flux binaires manipulent des octets sans conversion; utiles pour firmware, trames capteur et sommes de contrôle.

CODE / API EXAMPLE

```
from io import BytesIO
with BytesIO(b"\x01\x02") as f:
    print(f.read().hex())
```



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Chapter 186 - Reading Files

ENGLISH	FRANÇAIS
What does Reading Files actually do? Read in bounded chunks for large captures; distinguish end-of-file from transient short reads on sockets and devices.	**Quel est le rôle de Reading Files ?** Lire par blocs limités les gros fichiers; distinguer fin de fichier et lecture partielle sur un périphérique.
What is the engineering design constraint? Specify the external file contract and guard against truncation, platform differences and interruption.	**Quelle contrainte de conception faut-il respecter ?** Définir le contrat fichier et protéger contre troncature, différences OS et interruption.
How should the behavior be verified? Test missing, empty, truncated and read-only files as appropriate.	**Comment vérifier le comportement ?** Tester fichiers absents, vides, tronqués et accès refusé selon le cas.

ENGINEERING NOTE / NOTE TECHNIQUE

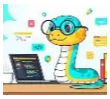
ENGINEERING NOTE / NOTE TECHNIQUE: Read in bounded chunks for large captures; distinguish end-of-file from transient short reads on sockets and devices. / Lire par blocs limités les gros fichiers; distinguer fin de fichier et lecture partielle sur un périphérique.

CODE / API EXAMPLE

```
from io import BytesIO
with BytesIO(b"abcdef") as f:
    for chunk in iter(lambda: f.read(2), b""):
        print(chunk)
```



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Chapter 187 - Writing Files

ENGLISH	FRANÇAIS
What does Writing Files actually do? Writes may be buffered; check exceptions, required byte counts where applicable and durability separately from successful write return.	**Quel est le rôle de Writing Files ?** Les écritures sont souvent tamponnées; vérifier erreurs, volumes réellement écrits et persistance séparément.
What is the engineering design constraint? Specify the external file contract and guard against truncation, platform differences and interruption.	**Quelle contrainte de conception faut-il respecter ?** Définir le contrat fichier et protéger contre troncature, différences OS et interruption.
How should the behavior be verified? Test missing, empty, truncated and read-only files as appropriate.	**Comment vérifier le comportement ?** Tester fichiers absents, vides, tronqués et accès refusé selon le cas.

ENGINEERING NOTE / NOTE TECHNIQUE

ENGINEERING NOTE / NOTE TECHNIQUE: Writes may be buffered; check exceptions, required byte counts where applicable and durability separately from successful write return. / Les écritures sont souvent tamponnées; vérifier erreurs, volumes réellement écrits et persistance séparément.

CODE / API EXAMPLE

```
from io import BytesIO
with BytesIO() as f:
   n = f.write(b"ABC")
   assert n == 3
   print(f.getvalue())
```



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Chapter 188 - Appending Data

ENGLISH	FRANÇAIS
What does Appending Data actually do? Append mode preserves existing content but does not by itself guarantee transactional records under concurrency or power loss.	**Quel est le rôle de Appending Data ?** Le mode ajout conserve le contenu mais ne garantit ni transaction ni intégrité après panne d'alimentation.
What is the engineering design constraint? Specify the external file contract and guard against truncation, platform differences and interruption.	**Quelle contrainte de conception faut-il respecter ?** Définir le contrat fichier et protéger contre troncature, différences OS et interruption.
How should the behavior be verified? Test missing, empty, truncated and read-only files as appropriate.	**Comment vérifier le comportement ?** Tester fichiers absents, vides, tronqués et accès refusé selon le cas.

ENGINEERING NOTE / NOTE TECHNIQUE

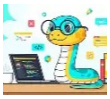
ENGINEERING NOTE / NOTE TECHNIQUE: Append mode preserves existing content but does not by itself guarantee transactional records under concurrency or power loss. / Le mode ajout conserve le contenu mais ne garantit ni transaction ni intégrité après panne d'alimentation.

CODE / API EXAMPLE

```
from tempfile import TemporaryFile
with TemporaryFile(mode="a+t", encoding="utf-8") as f:
    f.write("run1\n"); f.write("run2\n")
    f.seek(0); print(f.read())
```



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Chapter 189 - File Position - tell() & seek()

ENGLISH	FRANÇAIS
What does File Position - tell() & seek() actually do? tell reports stream position and seek repositions it; in text mode, opaque cookies should not be treated as raw byte offsets.	**Quel est le rôle de File Position - tell() & seek() ?** tell donne la position et seek la modifie; en mode texte les valeurs peuvent être des jetons opaques.
What is the engineering design constraint? Specify the external file contract and guard against truncation, platform differences and interruption.	**Quelle contrainte de conception faut-il respecter ?** Définir le contrat fichier et protéger contre troncature, différences OS et interruption.
How should the behavior be verified? Test missing, empty, truncated and read-only files as appropriate.	**Comment vérifier le comportement ?** Tester fichiers absents, vides, tronqués et accès refusé selon le cas.

ENGINEERING NOTE / NOTE TECHNIQUE

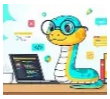
ENGINEERING NOTE / NOTE TECHNIQUE: tell reports stream position and seek repositions it; in text mode, opaque cookies should not be treated as raw byte offsets. / tell donne la position et seek la modifie; en mode texte les valeurs peuvent être des jetons opaques.

CODE / API EXAMPLE

```
from io import BytesIO
f = BytesIO(b"abcdef")
f.seek(3); print(f.tell(), f.read(2))
```



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Chapter 190 - Buffering

ENGLISH	FRANÇAIS
What does Buffering actually do? Buffering improves throughput but delays physical writes; flush requests propagation to the OS, while fsync is a separate durability step.	**Quel est le rôle de Buffering ?** Le tamponnement augmente le débit mais retarde l'écriture; flush et fsync répondent à des niveaux différents de persistance.
What is the engineering design constraint? Specify the external file contract and guard against truncation, platform differences and interruption.	**Quelle contrainte de conception faut-il respecter ?** Définir le contrat fichier et protéger contre troncature, différences OS et interruption.
How should the behavior be verified? Test missing, empty, truncated and read-only files as appropriate.	**Comment vérifier le comportement ?** Tester fichiers absents, vides, tronqués et accès refusé selon le cas.

ENGINEERING NOTE / NOTE TECHNIQUE

ENGINEERING NOTE / NOTE TECHNIQUE: Buffering improves throughput but delays physical writes; flush requests propagation to the OS, while fsync is a separate durability step. / Le tamponnement augmente le débit mais retarde l'écriture; flush et fsync répondent à des niveaux différents de persistance.

CODE / API EXAMPLE

```
from io import StringIO
f = StringIO(); f.write("sample\n"); f.flush(); print(f.getvalue())
```



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Chapter 191 - Text Encoding & Decoding

ENGLISH	FRANÇAIS
What does Text Encoding & Decoding actually do? Explicit decode errors indicate an invalid encoding or damaged data; do not silently replace corrupted bytes in measurement records.	**Quel est le rôle de Text Encoding & Decoding ?** Une erreur de décodage signale un encodage ou des données incorrects; éviter le remplacement silencieux dans les mesures.
What is the engineering design constraint? Specify the external file contract and guard against truncation, platform differences and interruption.	**Quelle contrainte de conception faut-il respecter ?** Définir le contrat fichier et protéger contre troncature, différences OS et interruption.
How should the behavior be verified? Test missing, empty, truncated and read-only files as appropriate.	**Comment vérifier le comportement ?** Tester fichiers absents, vides, tronqués et accès refusé selon le cas.

ENGINEERING NOTE / NOTE TECHNIQUE

ENGINEERING NOTE / NOTE TECHNIQUE: Explicit decode errors indicate an invalid encoding or damaged data; do not silently replace corrupted bytes in measurement records. / Une erreur de décodage signale un encodage ou des données incorrects; éviter le remplacement silencieux dans les mesures.

CODE / API EXAMPLE

```
payload = "étalon".encode("utf-8")
print(payload.decode("utf-8"))
```



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Chapter 192 - UTF-8 Strategy

ENGLISH	FRANÇAIS
What does UTF-8 Strategy actually do? UTF-8 is a portable default for text logs and metadata; document normalization and BOM policy when exchanging files with external tools.	**Quel est le rôle de UTF-8 Strategy ?** UTF-8 est un bon défaut pour journaux et métadonnées; documenter normalisation et BOM lors des échanges.
What is the engineering design constraint? Specify the external file contract and guard against truncation, platform differences and interruption.	**Quelle contrainte de conception faut-il respecter ?** Définir le contrat fichier et protéger contre troncature, différences OS et interruption.
How should the behavior be verified? Test missing, empty, truncated and read-only files as appropriate.	**Comment vérifier le comportement ?** Tester fichiers absents, vides, tronqués et accès refusé selon le cas.

ENGINEERING NOTE / NOTE TECHNIQUE

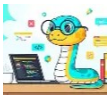
ENGINEERING NOTE / NOTE TECHNIQUE: UTF-8 is a portable default for text logs and metadata; document normalization and BOM policy when exchanging files with external tools. / UTF-8 est un bon défaut pour journaux et métadonnées; documenter normalisation et BOM lors des échanges.

CODE / API EXAMPLE

```
from pathlib import Path
p = Path("unicode.txt")
p.write_text("résistance", encoding="utf-8")
assert p.read_text(encoding="utf-8") == "résistance"
```



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Chapter 193 - Newline Handling

ENGLISH	FRANÇAIS
What does Newline Handling actually do? newline conversion varies by platform in text mode; use explicit newline for machine-readable formats such as CSV.	**Quel est le rôle de Newline Handling ?** La conversion des fins de ligne dépend du système; choisir newline explicitement pour les formats échangés.
What is the engineering design constraint? Specify the external file contract and guard against truncation, platform differences and interruption.	**Quelle contrainte de conception faut-il respecter ?** Définir le contrat fichier et protéger contre troncature, différences OS et interruption.
How should the behavior be verified? Test missing, empty, truncated and read-only files as appropriate.	**Comment vérifier le comportement ?** Tester fichiers absents, vides, tronqués et accès refusé selon le cas.

ENGINEERING NOTE / NOTE TECHNIQUE

ENGINEERING NOTE / NOTE TECHNIQUE: newline conversion varies by platform in text mode; use explicit newline for machine-readable formats such as CSV. / La conversion des fins de ligne dépend du système; choisir newline explicitement pour les formats échangés.

CODE / API EXAMPLE

```
import csv, io
buf = io.StringIO(newline="")
csv.writer(buf).writerow(["channel", "value"])
print(repr(buf.getvalue()))
```



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Chapter 194 - Large Files & Streaming

ENGLISH	FRANÇAIS
What does Large Files & Streaming actually do? Streaming reduces memory use and permits bounded processing; define chunk size, backpressure and behavior for malformed or truncated records.	**Quel est le rôle de Large Files & Streaming ?** Le traitement par flux limite la mémoire; fixer taille de bloc, pression aval et réaction aux trames incomplètes.
What is the engineering design constraint? Specify the external file contract and guard against truncation, platform differences and interruption.	**Quelle contrainte de conception faut-il respecter ?** Définir le contrat fichier et protéger contre troncature, différences OS et interruption.
How should the behavior be verified? Test missing, empty, truncated and read-only files as appropriate.	**Comment vérifier le comportement ?** Tester fichiers absents, vides, tronqués et accès refusé selon le cas.

ENGINEERING NOTE / NOTE TECHNIQUE

ENGINEERING NOTE / NOTE TECHNIQUE: Streaming reduces memory use and permits bounded processing; define chunk size, backpressure and behavior for malformed or truncated records. / Le traitement par flux limite la mémoire; fixer taille de bloc, pression aval et réaction aux trames incomplètes.

CODE / API EXAMPLE

```
from io import BytesIO
src = BytesIO(b"0123456789")
total = 0
for block in iter(lambda: src.read(4), b""):
    total += len(block)
print(total)
```



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Chapter 195 - pathlib

ENGLISH	FRANÇAIS
What does pathlib actually do? pathlib represents paths without string concatenation; relative paths, symlinks and platform-specific semantics still require care.	**Quel est le rôle de pathlib ?** pathlib manipule les chemins sans concaténation manuelle; chemins relatifs et liens symboliques exigent une validation.
What is the engineering design constraint? Specify the external file contract and guard against truncation, platform differences and interruption.	**Quelle contrainte de conception faut-il respecter ?** Définir le contrat fichier et protéger contre troncature, différences OS et interruption.
How should the behavior be verified? Test missing, empty, truncated and read-only files as appropriate.	**Comment vérifier le comportement ?** Tester fichiers absents, vides, tronqués et accès refusé selon le cas.

ENGINEERING NOTE / NOTE TECHNIQUE

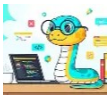
ENGINEERING NOTE / NOTE TECHNIQUE: pathlib represents paths without string concatenation; relative paths, symlinks and platform-specific semantics still require care. / pathlib manipule les chemins sans concaténation manuelle; chemins relatifs et liens symboliques exigent une validation.

CODE / API EXAMPLE

```
from pathlib import Path
p = Path("logs") / "run_001.csv"
print(p.suffix, p.name)
```



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Chapter 196 - Directory Traversal

ENGLISH	FRANÇAIS
What does Directory Traversal actually do? Directory traversal may include unexpected symlinks and huge trees; bound scope, handle permission errors and never assume filesystem order.	**Quel est le rôle de Directory Traversal ?** Le parcours d'arborescence peut suivre des liens et parcourir un grand volume; limiter l'étendue et gérer les permissions.
What is the engineering design constraint? Specify the external file contract and guard against truncation, platform differences and interruption.	**Quelle contrainte de conception faut-il respecter ?** Définir le contrat fichier et protéger contre troncature, différences OS et interruption.
How should the behavior be verified? Test missing, empty, truncated and read-only files as appropriate.	**Comment vérifier le comportement ?** Tester fichiers absents, vides, tronqués et accès refusé selon le cas.

ENGINEERING NOTE / NOTE TECHNIQUE

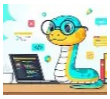
ENGINEERING NOTE / NOTE TECHNIQUE: Directory traversal may include unexpected symlinks and huge trees; bound scope, handle permission errors and never assume filesystem order. / Le parcours d'arborescence peut suivre des liens et parcourir un grand volume; limiter l'étendue et gérer les permissions.

CODE / API EXAMPLE

```
from pathlib import Path
for child in sorted(Path(".").iterdir(), key=lambda p:p.name)[:3]:
    print(child.name)
```



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Chapter 197 - File Metadata

ENGLISH	FRANÇAIS
What does File Metadata actually do? File metadata includes size, type and modification time; timestamps are not reliable content identity or cryptographic evidence.	**Quel est le rôle de File Metadata ?** Les métadonnées comprennent taille, type et date; elles ne prouvent pas l'identité cryptographique d'un fichier.
What is the engineering design constraint? Specify the external file contract and guard against truncation, platform differences and interruption.	**Quelle contrainte de conception faut-il respecter ?** Définir le contrat fichier et protéger contre troncature, différences OS et interruption.
How should the behavior be verified? Test missing, empty, truncated and read-only files as appropriate.	**Comment vérifier le comportement ?** Tester fichiers absents, vides, tronqués et accès refusé selon le cas.

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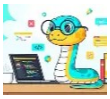
ENGINEERING NOTE / NOTE TECHNIQUE: File metadata includes size, type and modification time; timestamps are not reliable content identity or cryptographic evidence. / Les métadonnées comprennent taille, type et date; elles ne prouvent pas l'identité cryptographique d'un fichier.

CODE / API EXAMPLE

```
from pathlib import Path
p = Path(".")
s = p.stat()
print(s.st_size, s.st_mtime_ns)
```



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Chapter 198 - Permissions Concepts

ENGLISH	FRANÇAIS
What does Permissions Concepts actually do? Permissions differ across Windows and POSIX; use least privilege but verify ACLs, ownership and inherited rules beyond simple mode bits.	**Quel est le rôle de Permissions Concepts ?** Les droits diffèrent sous Windows et POSIX; privilégier les permissions minimales et vérifier ACL et héritage.
What is the engineering design constraint? Specify the external file contract and guard against truncation, platform differences and interruption.	**Quelle contrainte de conception faut-il respecter ?** Définir le contrat fichier et protéger contre troncature, différences OS et interruption.
How should the behavior be verified? Test missing, empty, truncated and read-only files as appropriate.	**Comment vérifier le comportement ?** Tester fichiers absents, vides, tronqués et accès refusé selon le cas.

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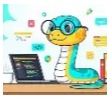
ENGINEERING NOTE / NOTE TECHNIQUE: Permissions differ across Windows and POSIX; use least privilege but verify ACLs, ownership and inherited rules beyond simple mode bits. / Les droits diffèrent sous Windows et POSIX; privilégier les permissions minimales et vérifier ACL et héritage.

CODE / API EXAMPLE

```
import os, stat
mode = stat.S_IMODE(os.stat(".").st_mode)
print(oct(mode))
```



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Chapter 199 - shutil & File Operations

ENGLISH	FRANÇAIS
What does shutil & File Operations actually do? shutil copies files and directories but metadata preservation is platform dependent; verify destination and handle partial copy failures.	**Quel est le rôle de shutil & File Operations ?** shutil copie des fichiers et dossiers; la conservation des attributs varie selon le système et les copies peuvent être partielles.
What is the engineering design constraint? Specify the external file contract and guard against truncation, platform differences and interruption.	**Quelle contrainte de conception faut-il respecter ?** Définir le contrat fichier et protéger contre troncature, différences OS et interruption.
How should the behavior be verified? Test missing, empty, truncated and read-only files as appropriate.	**Comment vérifier le comportement ?** Tester fichiers absents, vides, tronqués et accès refusé selon le cas.

ENGINEERING NOTE / NOTE TECHNIQUE

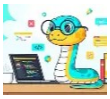
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CODE / API EXAMPLE

```
from pathlib import Path
import shutil, tempfile
with tempfile.TemporaryDirectory() as folder:
    a=Path(folder)/"a"; a.write_bytes(b"data")
    b=Path(folder)/"b"; shutil.copyfile(a,b)
    print(b.read_bytes())
```



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Chapter 200 - Temporary Files & Directories

ENGLISH	FRANÇAIS
What does Temporary Files & Directories actually do? TemporaryFile and TemporaryDirectory handle cleanup; secure workflows must also protect filenames, directory permissions and crash leftovers.	**Quel est le rôle de Temporary Files & Directories ?** TemporaryFile et TemporaryDirectory simplifient le nettoyage; vérifier aussi droits, noms et résidus après crash.
What is the engineering design constraint? Specify the external file contract and guard against truncation, platform differences and interruption.	**Quelle contrainte de conception faut-il respecter ?** Définir le contrat fichier et protéger contre troncature, différences OS et interruption.
How should the behavior be verified? Test missing, empty, truncated and read-only files as appropriate.	**Comment vérifier le comportement ?** Tester fichiers absents, vides, tronqués et accès refusé selon le cas.

ENGINEERING NOTE / NOTE TECHNIQUE

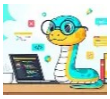
ENGINEERING NOTE / NOTE TECHNIQUE: TemporaryFile and TemporaryDirectory handle cleanup; secure workflows must also protect filenames, directory permissions and crash leftovers. / TemporaryFile et TemporaryDirectory simplifient le nettoyage; vérifier aussi droits, noms et résidus après crash.

CODE / API EXAMPLE

```
from tempfile import TemporaryDirectory
from pathlib import Path
with TemporaryDirectory() as folder:
    p=Path(folder)/"capture.bin"
    p.write_bytes(b"OK")
    print(p.exists())
```



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Chapter 201 - Safe & Atomic File Replacement

ENGLISH	FRANÇAIS
What does Safe & Atomic File Replacement actually do? A same-filesystem os.replace offers atomic name replacement, not guaranteed power-loss durability; flush and fsync file and parent directory where supported.	**Quel est le rôle de Safe & Atomic File Replacement ?** os.replace remplace atomiquement le nom sur un même système de fichiers, mais la durabilité exige des synchronisations adaptées.
What is the engineering design constraint? Specify the external file contract and guard against truncation, platform differences and interruption.	**Quelle contrainte de conception faut-il respecter ?** Définir le contrat fichier et protéger contre troncature, différences OS et interruption.
How should the behavior be verified? Test missing, empty, truncated and read-only files as appropriate.	**Comment vérifier le comportement ?** Tester fichiers absents, vides, tronqués et accès refusé selon le cas.

ENGINEERING NOTE / NOTE TECHNIQUE

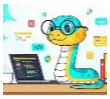
ENGINEERING NOTE / NOTE TECHNIQUE: A same-filesystem os.replace offers atomic name replacement, not guaranteed power-loss durability; flush and fsync file and parent directory where supported. / os.replace remplace atomiquement le nom sur un même système de fichiers, mais la durabilité exige des synchronisations adaptées.

CODE / API EXAMPLE

```
import os, tempfile
from pathlib import Path
with tempfile.TemporaryDirectory() as d:
    target=Path(d)/"state"; tmp=Path(d)/"state.tmp"
    tmp.write_bytes(b"v2")
    os.replace(tmp, target)
    print(target.read_bytes())
```



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Chapter 202 - Date & Time Fundamentals

ENGLISH	FRANÇAIS
What does Date & Time Fundamentals actually do? Wall-clock time labels events, while monotonic clocks measure elapsed durations; clocks may jump due to synchronization or manual changes.	**Quel est le rôle de Date & Time Fundamentals ?** L'horloge civile date les événements; une horloge monotone mesure les durées malgré les corrections d'heure.
What is the engineering design constraint? State the clock type, time zone and precision required by each measurement.	**Quelle contrainte de conception faut-il respecter ?** Déclarer horloge, fuseau et précision de chaque mesure.
How should the behavior be verified? Test clock jumps, timezone changes and ordering of near-simultaneous records.	**Comment vérifier le comportement ?** Tester changement d'heure, de fuseau et ordre des acquisitions.

ENGINEERING NOTE / NOTE TECHNIQUE

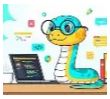
ENGINEERING NOTE / NOTE TECHNIQUE: Wall-clock time labels events, while monotonic clocks measure elapsed durations; clocks may jump due to synchronization or manual changes. / L'horloge civile date les événements; une horloge monotone mesure les durées malgré les corrections d'heure.

CODE / API EXAMPLE

```
import time
start=time.monotonic_ns()
work=sum(range(1000))
print(time.monotonic_ns() >= start)
```



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Chapter 203 - datetime, date, time & timedelta

ENGLISH	FRANÇAIS
What does datetime, date, time & timedelta actually do? datetime/date/time model calendar values, while timedelta is a duration; naive and aware datetimes must not be mixed casually.	**Quel est le rôle de datetime, date, time & timedelta ?** datetime/date/time modélisent le calendrier et timedelta une durée; distinguer les heures avec ou sans fuseau.
What is the engineering design constraint? State the clock type, time zone and precision required by each measurement.	**Quelle contrainte de conception faut-il respecter ?** Déclarer horloge, fuseau et précision de chaque mesure.
How should the behavior be verified? Test clock jumps, timezone changes and ordering of near-simultaneous records.	**Comment vérifier le comportement ?** Tester changement d'heure, de fuseau et ordre des acquisitions.

ENGINEERING NOTE / NOTE TECHNIQUE

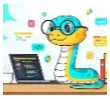
ENGINEERING NOTE / NOTE TECHNIQUE: datetime/date/time model calendar values, while timedelta is a duration; naive and aware datetimes must not be mixed casually. / datetime/date/time modélisent le calendrier et timedelta une durée; distinguer les heures avec ou sans fuseau.

CODE / API EXAMPLE

```
from datetime import datetime, timedelta, timezone
now=datetime.now(timezone.utc)
print(now + timedelta(seconds=5) > now)
```



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Chapter 204 - UTC, Time Zones & zoneinfo

ENGLISH	FRANÇAIS
What does UTC, Time Zones & zoneinfo actually do? Use timezone-aware UTC instants for storage and zoneinfo for local presentation; daylight-saving transitions can create ambiguous times.	**Quel est le rôle de UTC, Time Zones & zoneinfo ?** Stocker les instants en UTC avec fuseau; utiliser zoneinfo pour l'affichage, en tenant compte des transitions été/hiver.
What is the engineering design constraint? State the clock type, time zone and precision required by each measurement.	**Quelle contrainte de conception faut-il respecter ?** Déclarer horloge, fuseau et précision de chaque mesure.
How should the behavior be verified? Test clock jumps, timezone changes and ordering of near-simultaneous records.	**Comment vérifier le comportement ?** Tester changement d'heure, de fuseau et ordre des acquisitions.

ENGINEERING NOTE / NOTE TECHNIQUE

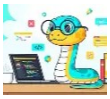
ENGINEERING NOTE / NOTE TECHNIQUE: Use timezone-aware UTC instants for storage and zoneinfo for local presentation; daylight-saving transitions can create ambiguous times. / Stocker les instants en UTC avec fuseau; utiliser zoneinfo pour l'affichage, en tenant compte des transitions été/hiver.

CODE / API EXAMPLE

```
from datetime import datetime, timezone
from zoneinfo import ZoneInfo
now=datetime.now(timezone.utc)
print(now.astimezone(ZoneInfo("America/Toronto")).utcoffset())
```



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Chapter 205 - Engineering Timestamps & Time Conversion

ENGLISH	FRANÇAIS
What does Engineering Timestamps & Time Conversion actually do? Acquisition timestamps need declared clock source, resolution and synchronization; sequence numbers distinguish lost records even with stable timestamps.	**Quel est le rôle de Engineering Timestamps & Time Conversion ?** Un horodatage de mesure exige source, résolution et synchronisation définies; les numéros de séquence révèlent les pertes.
What is the engineering design constraint? State the clock type, time zone and precision required by each measurement.	**Quelle contrainte de conception faut-il respecter ?** Déclarer horloge, fuseau et précision de chaque mesure.
How should the behavior be verified? Test clock jumps, timezone changes and ordering of near-simultaneous records.	**Comment vérifier le comportement ?** Tester changement d'heure, de fuseau et ordre des acquisitions.

ENGINEERING NOTE / NOTE TECHNIQUE

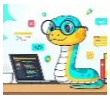
ENGINEERING NOTE / NOTE TECHNIQUE: Acquisition timestamps need declared clock source, resolution and synchronization; sequence numbers distinguish lost records even with stable timestamps. / Un horodatage de mesure exige source, résolution et synchronisation définies; les numéros de séquence révèlent les pertes.

CODE / API EXAMPLE

```
import time
sample={"seq":7,"t_mono_ns":time.monotonic_ns(),"value":1.2}
print(sample["seq"])
```



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Chapter 206 - CSV & Engineering Logs

ENGLISH	FRANÇAIS
What does CSV & Engineering Logs actually do? CSV is interoperable but lacks native types; define column names, units, delimiter, quoting and locale-independent numeric representation.	**Quel est le rôle de CSV & Engineering Logs ?** CSV facilite l'échange mais ne stocke pas les types; préciser unités, colonnes, séparateur et convention décimale.
What is the engineering design constraint? Version the format and reject records violating length, type, unit or integrity constraints.	**Quelle contrainte de conception faut-il respecter ?** Versionner le format et refuser les enregistrements mal typés ou incomplets.
How should the behavior be verified? Round-trip representative and malformed records, including boundary values.	**Comment vérifier le comportement ?** Tester aller-retour, données mal formées et valeurs limites.

ENGINEERING NOTE / NOTE TECHNIQUE

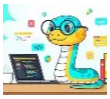
ENGINEERING NOTE / NOTE TECHNIQUE: CSV is interoperable but lacks native types; define column names, units, delimiter, quoting and locale-independent numeric representation. / CSV facilite l'échange mais ne stocke pas les types; préciser unités, colonnes, séparateur et convention décimale.

CODE / API EXAMPLE

```
import csv, io
out=io.StringIO(newline="")
w=csv.writer(out); w.writerow(["time_s","voltage_V"]); w.writerow([0.1,3.3])
print(out.getvalue())
```



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Chapter 207 - JSON

ENGLISH	FRANÇAIS
What does JSON actually do? JSON supports portable structured configuration and telemetry but not comments, NaN by standard or exact binary floats; validate schema after parsing.	**Quel est le rôle de JSON ?** JSON transporte des structures mais ne garantit ni schéma ni précision des flottants; valider après lecture.
What is the engineering design constraint? Version the format and reject records violating length, type, unit or integrity constraints.	**Quelle contrainte de conception faut-il respecter ?** Versionner le format et refuser les enregistrements mal typés ou incomplets.
How should the behavior be verified? Round-trip representative and malformed records, including boundary values.	**Comment vérifier le comportement ?** Tester aller-retour, données mal formées et valeurs limites.

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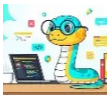
ENGINEERING NOTE / NOTE TECHNIQUE: JSON supports portable structured configuration and telemetry but not comments, NaN by standard or exact binary floats; validate schema after parsing. / JSON transporte des structures mais ne garantit ni schéma ni précision des flottants; valider après lecture.

CODE / API EXAMPLE

```
import json
obj=json.loads('{"channel":2,"enabled":true}')
assert isinstance(obj["channel"],int)
print(obj)
```



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Chapter 208 - TOML

ENGLISH	FRANÇAIS
What does TOML actually do? TOML is useful for human-edited configuration; Python 3.11+ provides tomllib for reading, while writing requires another library.	**Quel est le rôle de TOML ?** TOML convient aux configurations lisibles; tomllib (Python 3.11+) lit les fichiers mais ne les écrit pas.
What is the engineering design constraint? Version the format and reject records violating length, type, unit or integrity constraints.	**Quelle contrainte de conception faut-il respecter ?** Versionner le format et refuser les enregistrements mal typés ou incomplets.
How should the behavior be verified? Round-trip representative and malformed records, including boundary values.	**Comment vérifier le comportement ?** Tester aller-retour, données mal formées et valeurs limites.

ENGINEERING NOTE / NOTE TECHNIQUE

ENGINEERING NOTE / NOTE TECHNIQUE: TOML is useful for human-edited configuration; Python 3.11+ provides tomllib for reading, while writing requires another library. / TOML convient aux configurations lisibles; tomllib (Python 3.11+) lit les fichiers mais ne les écrit pas.

CODE / API EXAMPLE

```
import tomllib
settings=tomllib.loads('sample_rate = 1000\n[adc]\nbits = 12')
print(settings["adc"]["bits"])
```



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Chapter 209 - INI / Configuration Formats

ENGLISH	FRANÇAIS
What does INI / Configuration Formats actually do? INI files are simple but preserve most values as strings; ConfigParser interpolation and case-folding need explicit policies.	**Quel est le rôle de INI / Configuration Formats ?** INI reste simple, mais les valeurs sont surtout des chaînes; documenter interpolation et sensibilité à la casse.
What is the engineering design constraint? Version the format and reject records violating length, type, unit or integrity constraints.	**Quelle contrainte de conception faut-il respecter ?** Versionner le format et refuser les enregistrements mal typés ou incomplets.
How should the behavior be verified? Round-trip representative and malformed records, including boundary values.	**Comment vérifier le comportement ?** Tester aller-retour, données mal formées et valeurs limites.

ENGINEERING NOTE / NOTE TECHNIQUE

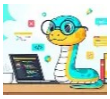
ENGINEERING NOTE / NOTE TECHNIQUE: INI files are simple but preserve most values as strings; ConfigParser interpolation and case-folding need explicit policies. / INI reste simple, mais les valeurs sont surtout des chaînes; documenter interpolation et sensibilité à la casse.

CODE / API EXAMPLE

```
from configparser import ConfigParser
cfg=ConfigParser(); cfg.read_string('[daq]\nrate=1000')
print(cfg.getint('daq','rate'))
```



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Chapter 210 - XML

ENGLISH	FRANÇAIS
What does XML actually do? XML adds attributes and namespaces; parse untrusted XML with hardened parsers and bound resource use according to deployment threat model.	**Quel est le rôle de XML ?** XML comporte attributs et espaces de noms; durcir l'analyse des documents non fiables et limiter les ressources.
What is the engineering design constraint? Version the format and reject records violating length, type, unit or integrity constraints.	**Quelle contrainte de conception faut-il respecter ?** Versionner le format et refuser les enregistrements mal typés ou incomplets.
How should the behavior be verified? Round-trip representative and malformed records, including boundary values.	**Comment vérifier le comportement ?** Tester aller-retour, données mal formées et valeurs limites.

ENGINEERING NOTE / NOTE TECHNIQUE

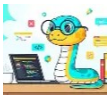
ENGINEERING NOTE / NOTE TECHNIQUE: XML adds attributes and namespaces; parse untrusted XML with hardened parsers and bound resource use according to deployment threat model. / XML comporte attributs et espaces de noms; durcir l'analyse des documents non fiables et limiter les ressources.

CODE / API EXAMPLE

```
from xml.etree import ElementTree as ET
root=ET.fromstring('<record channel="2">3.3</record>')
print(root.attrib["channel"], root.text)
```



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Chapter 211 - Binary Data Formats

ENGLISH	FRANÇAIS
What does Binary Data Formats actually do? A binary format requires explicit magic bytes, version, length, units, byte order and integrity checks; never cast arbitrary bytes to trusted structures.	**Quel est le rôle de Binary Data Formats ?** Un format binaire doit documenter signature, version, longueur, unités, ordre des octets et intégrité.
What is the engineering design constraint? Version the format and reject records violating length, type, unit or integrity constraints.	**Quelle contrainte de conception faut-il respecter ?** Versionner le format et refuser les enregistrements mal typés ou incomplets.
How should the behavior be verified? Round-trip representative and malformed records, including boundary values.	**Comment vérifier le comportement ?** Tester aller-retour, données mal formées et valeurs limites.

ENGINEERING NOTE / NOTE TECHNIQUE

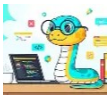
ENGINEERING NOTE / NOTE TECHNIQUE: A binary format requires explicit magic bytes, version, length, units, byte order and integrity checks; never cast arbitrary bytes to trusted structures. / Un format binaire doit documenter signature, version, longueur, unités, ordre des octets et intégrité.

CODE / API EXAMPLE

```
header=b"AHDS"+bytes([1, 4])
assert header[:4]==b"AHDS" and header[4]==1
print(header.hex())
```



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Chapter 212 - struct Introduction

ENGLISH	FRANÇAIS
What does struct Introduction actually do? struct.pack/unpack map defined layouts to bytes; always specify a prefix such as '<' or '>' to avoid native alignment surprises.	**Quel est le rôle de struct Introduction ?** struct.pack et unpack sérialisent des champs; imposer '<' ou '>' pour éviter alignement et ordre natifs.
What is the engineering design constraint? Version the format and reject records violating length, type, unit or integrity constraints.	**Quelle contrainte de conception faut-il respecter ?** Versionner le format et refuser les enregistrements mal typés ou incomplets.
How should the behavior be verified? Round-trip representative and malformed records, including boundary values.	**Comment vérifier le comportement ?** Tester aller-retour, données mal formées et valeurs limites.

ENGINEERING NOTE / NOTE TECHNIQUE

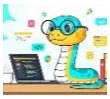
ENGINEERING NOTE / NOTE TECHNIQUE: struct.pack/unpack map defined layouts to bytes; always specify a prefix such as '<' or '>' to avoid native alignment surprises. / struct.pack et unpack sérialisent des champs; imposer '<' ou '>' pour éviter alignement et ordre natifs.

CODE / API EXAMPLE

```
import struct
packet=struct.pack('<Hf',0x1234,1.5)
print(struct.unpack('<Hf',packet))
```



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Chapter 213 - Endianness in File Formats - Introduction

ENGLISH	FRANÇAIS
What does Endianness in File Formats - Introduction actually do? Little- and big-endian encodings change multi-byte interpretation; protocol specification, not host CPU, determines order.	**Quel est le rôle de Endianness in File Formats - Introduction ?** Les ordres little-endian et big-endian changent la valeur décodée; suivre la spécification du protocole.
What is the engineering design constraint? Version the format and reject records violating length, type, unit or integrity constraints.	**Quelle contrainte de conception faut-il respecter ?** Versionner le format et refuser les enregistrements mal typés ou incomplets.
How should the behavior be verified? Round-trip representative and malformed records, including boundary values.	**Comment vérifier le comportement ?** Tester aller-retour, données mal formées et valeurs limites.

ENGINEERING NOTE / NOTE TECHNIQUE

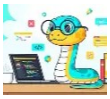
ENGINEERING NOTE / NOTE TECHNIQUE: Little- and big-endian encodings change multi-byte interpretation; protocol specification, not host CPU, determines order. / Les ordres little-endian et big-endian changent la valeur décodée; suivre la spécification du protocole.

CODE / API EXAMPLE

```
word=0x1234
print(word.to_bytes(2,'little').hex(),word.to_bytes(2,'big').hex())
```



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Chapter 214 - Fixed-Width Binary Records

ENGLISH	FRANÇAIS
What does Fixed-Width Binary Records actually do? Fixed records require a documented layout and precise size; detect truncation and avoid relying on platform-native structure padding.	**Quel est le rôle de Fixed-Width Binary Records ?** Les enregistrements fixes exigent taille et disposition définies; refuser les données tronquées et l'alignement implicite.
What is the engineering design constraint? Version the format and reject records violating length, type, unit or integrity constraints.	**Quelle contrainte de conception faut-il respecter ?** Versionner le format et refuser les enregistrements mal typés ou incomplets.
How should the behavior be verified? Round-trip representative and malformed records, including boundary values.	**Comment vérifier le comportement ?** Tester aller-retour, données mal formées et valeurs limites.

ENGINEERING NOTE / NOTE TECHNIQUE

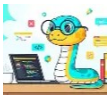
ENGINEERING NOTE / NOTE TECHNIQUE: Fixed records require a documented layout and precise size; detect truncation and avoid relying on platform-native structure padding. / Les enregistrements fixes exigent taille et disposition définies; refuser les données tronquées et l'alignement implicite.

CODE / API EXAMPLE

```
import struct
record=struct.pack('<Ih',17,-2)
assert len(record)==struct.calcsize('<Ih')
print(struct.unpack('<Ih',record))
```



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Chapter 215 - Binary File Parsing

ENGLISH	FRANÇAIS
What does Binary File Parsing actually do? Validate input length before unpacking; reject inconsistent version, declared lengths and checksums before using sensor values.	**Quel est le rôle de Binary File Parsing ?** Vérifier longueur, version et intégrité avant le décodage binaire et toute utilisation des mesures.
What is the engineering design constraint? Version the format and reject records violating length, type, unit or integrity constraints.	**Quelle contrainte de conception faut-il respecter ?** Versionner le format et refuser les enregistrements mal typés ou incomplets.
How should the behavior be verified? Round-trip representative and malformed records, including boundary values.	**Comment vérifier le comportement ?** Tester aller-retour, données mal formées et valeurs limites.

ENGINEERING NOTE / NOTE TECHNIQUE

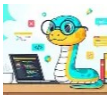
ENGINEERING NOTE / NOTE TECHNIQUE: Validate input length before unpacking; reject inconsistent version, declared lengths and checksums before using sensor values. / Vérifier longueur, version et intégrité avant le décodage binaire et toute utilisation des mesures.

CODE / API EXAMPLE

```
import struct
raw=b"\x01\x00\x2a\x00"
if len(raw)!=4: raise ValueError("truncated")
version,value=struct.unpack('<HH',raw)
print(version,value)
```



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Chapter 216 - Serialization Fundamentals

ENGLISH	FRANÇAIS
What does Serialization Fundamentals actually do? Serialization converts typed state into a portable representation; design version migration, validation and recovery from partial writes.	**Quel est le rôle de Serialization Fundamentals ?** La sérialisation convertit l'état en format stockable; prévoir version, validation et reprise après écriture incomplète.
What is the engineering design constraint? Version the format and reject records violating length, type, unit or integrity constraints.	**Quelle contrainte de conception faut-il respecter ?** Versionner le format et refuser les enregistrements mal typés ou incomplets.
How should the behavior be verified? Round-trip representative and malformed records, including boundary values.	**Comment vérifier le comportement ?** Tester aller-retour, données mal formées et valeurs limites.

ENGINEERING NOTE / NOTE TECHNIQUE

ENGINEERING NOTE / NOTE TECHNIQUE: Serialization converts typed state into a portable representation; design version migration, validation and recovery from partial writes. / La sérialisation convertit l'état en format stockable; prévoir version, validation et reprise après écriture incomplète.

CODE / API EXAMPLE

```
import json
payload=json.dumps({"v":1,"gain":2.5},allow_nan=False)
print(json.loads(payload)["gain"])
```



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Chapter 217 - pickle

ENGLISH	FRANÇAIS
What does pickle actually do? pickle can serialize complex Python objects but executes code when loading malicious payloads; use only with fully trusted data.	**Quel est le rôle de pickle ?** pickle restaure des objets Python mais un chargement malveillant peut exécuter du code; l'éviter pour toute source non fiable.
What is the engineering design constraint? Version the format and reject records violating length, type, unit or integrity constraints.	**Quelle contrainte de conception faut-il respecter ?** Versionner le format et refuser les enregistrements mal typés ou incomplets.
How should the behavior be verified? Round-trip representative and malformed records, including boundary values.	**Comment vérifier le comportement ?** Tester aller-retour, données mal formées et valeurs limites.

ENGINEERING NOTE / NOTE TECHNIQUE

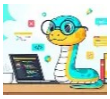
ENGINEERING NOTE / NOTE TECHNIQUE: pickle can serialize complex Python objects but executes code when loading malicious payloads; use only with fully trusted data. / pickle restaure des objets Python mais un chargement malveillant peut exécuter du code; l'éviter pour toute source non fiable.

CODE / API EXAMPLE

```
import pickle
trusted=pickle.dumps({"gain":1.5})
print(pickle.loads(trusted))  # ONLY trusted bytes
```



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Chapter 218 - Serialization Security

ENGLISH	FRANÇAIS
What does Serialization Security actually do? Untrusted deserialization is a code-execution boundary; prefer JSON/TOML plus strict schema checks over pickle for external sources.	**Quel est le rôle de Serialization Security ?** La désérialisation non fiable est une frontière d'exécution de code; préférer JSON/TOML et une validation stricte.
What is the engineering design constraint? Version the format and reject records violating length, type, unit or integrity constraints.	**Quelle contrainte de conception faut-il respecter ?** Versionner le format et refuser les enregistrements mal typés ou incomplets.
How should the behavior be verified? Round-trip representative and malformed records, including boundary values.	**Comment vérifier le comportement ?** Tester aller-retour, données mal formées et valeurs limites.

ENGINEERING NOTE / NOTE TECHNIQUE

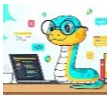
ENGINEERING NOTE / NOTE TECHNIQUE: Untrusted deserialization is a code-execution boundary; prefer JSON/TOML plus strict schema checks over pickle for external sources. / La désérialisation non fiable est une frontière d'exécution de code; préférer JSON/TOML et une validation stricte.

CODE / API EXAMPLE

```
import json
raw='{"gain":2.0}'
data=json.loads(raw)
if set(data)!={"gain"} or not isinstance(data["gain"],(int,float)):
    raise ValueError("invalid configuration")
```



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Chapter 219 - Data Validation

ENGLISH	FRANÇAIS
What does Data Validation actually do? Validation checks types, units, physical limits and cross-field relationships before any command reaches hardware.	**Quel est le rôle de Data Validation ?** La validation vérifie types, unités, limites physiques et relations entre champs avant de commander le matériel.
What is the engineering design constraint? Version the format and reject records violating length, type, unit or integrity constraints.	**Quelle contrainte de conception faut-il respecter ?** Versionner le format et refuser les enregistrements mal typés ou incomplets.
How should the behavior be verified? Round-trip representative and malformed records, including boundary values.	**Comment vérifier le comportement ?** Tester aller-retour, données mal formées et valeurs limites.

ENGINEERING NOTE / NOTE TECHNIQUE

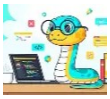
ENGINEERING NOTE / NOTE TECHNIQUE: Validation checks types, units, physical limits and cross-field relationships before any command reaches hardware. / La validation vérifie types, unités, limites physiques et relations entre champs avant de commander le matériel.

CODE / API EXAMPLE

```
def validate_adc(v):
   if isinstance(v,bool) or not isinstance(v,(int,float)) or not 0<=v<=3.0:
      raise ValueError("ADC volts outside limits")
   return float(v)
print(validate_adc(2.8))
```



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Chapter 220 - Configuration Validation

ENGLISH	FRANÇAIS
What does Configuration Validation actually do? Configuration validation must enforce permitted devices, rates, paths and fail-safe defaults; reject unknown unsafe fields.	**Quel est le rôle de Configuration Validation ?** Valider périphériques, fréquences et chemins autorisés; rejeter les paramètres dangereux ou inconnus.
What is the engineering design constraint? Version the format and reject records violating length, type, unit or integrity constraints.	**Quelle contrainte de conception faut-il respecter ?** Versionner le format et refuser les enregistrements mal typés ou incomplets.
How should the behavior be verified? Round-trip representative and malformed records, including boundary values.	**Comment vérifier le comportement ?** Tester aller-retour, données mal formées et valeurs limites.

ENGINEERING NOTE / NOTE TECHNIQUE

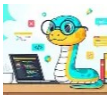
ENGINEERING NOTE / NOTE TECHNIQUE: Configuration validation must enforce permitted devices, rates, paths and fail-safe defaults; reject unknown unsafe fields. / Valider périphériques, fréquences et chemins autorisés; rejeter les paramètres dangereux ou inconnus.

CODE / API EXAMPLE

```
cfg={"rate_hz":1000,"channels":4}
if not 1<=cfg["channels"]<=16 or cfg["rate_hz"] not in (100,1000):
    raise ValueError("unsupported acquisition setup")
```



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Chapter 221 - Measurement Record Design

ENGLISH	FRANÇAIS
What does Measurement Record Design actually do? Measurement records need units, channel ID, timestamp provenance, calibration revision, status and validity flags.	**Quel est le rôle de Measurement Record Design ?** Un relevé doit contenir unités, voie, source temporelle, révision d'étalonnage et indicateur de validité.
What is the engineering design constraint? Version the format and reject records violating length, type, unit or integrity constraints.	**Quelle contrainte de conception faut-il respecter ?** Versionner le format et refuser les enregistrements mal typés ou incomplets.
How should the behavior be verified? Round-trip representative and malformed records, including boundary values.	**Comment vérifier le comportement ?** Tester aller-retour, données mal formées et valeurs limites.

ENGINEERING NOTE / NOTE TECHNIQUE

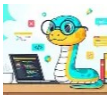
ENGINEERING NOTE / NOTE TECHNIQUE: Measurement records need units, channel ID, timestamp provenance, calibration revision, status and validity flags. / Un relevé doit contenir unités, voie, source temporelle, révision d'étalonnage et indicateur de validité.

CODE / API EXAMPLE

```
record={"channel":1,"value":2.5,"unit":"V","valid":True,"cal_rev":"B"}
assert record["unit"]=="V" and record["valid"]
```



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Chapter 222 - Engineering File Architecture

ENGLISH	FRANÇAIS
What does Engineering File Architecture actually do? Separate raw captures, derived results, configuration and audit metadata; version file schemas and make writes recoverable.	**Quel est le rôle de Engineering File Architecture ?** Séparer captures brutes, résultats, configuration et audit; versionner les formats et prévoir une reprise.
What is the engineering design constraint? Version the format and reject records violating length, type, unit or integrity constraints.	**Quelle contrainte de conception faut-il respecter ?** Versionner le format et refuser les enregistrements mal typés ou incomplets.
How should the behavior be verified? Round-trip representative and malformed records, including boundary values.	**Comment vérifier le comportement ?** Tester aller-retour, données mal formées et valeurs limites.

ENGINEERING NOTE / NOTE TECHNIQUE

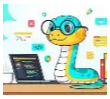
ENGINEERING NOTE / NOTE TECHNIQUE: Separate raw captures, derived results, configuration and audit metadata; version file schemas and make writes recoverable. / Séparer captures brutes, résultats, configuration et audit; versionner les formats et prévoir une reprise.

CODE / API EXAMPLE

```
from pathlib import Path
root=Path("test_run")
for name in ("raw","processed","metadata"):
    print(root/name)
```



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Chapter 223 - UUIDs & Unique Engineering Identifiers

ENGLISH	FRANÇAIS
What does UUIDs & Unique Engineering Identifiers actually do? UUIDs identify tests or acquisitions without central coordination, but they do not prove integrity, ordering or authenticity.	**Quel est le rôle de UUIDs & Unique Engineering Identifiers ?** Les UUID identifient les essais sans coordination centrale, mais ne prouvent ni intégrité, ni ordre, ni authenticité.
What is the engineering design constraint? Version the format and reject records violating length, type, unit or integrity constraints.	**Quelle contrainte de conception faut-il respecter ?** Versionner le format et refuser les enregistrements mal typés ou incomplets.
How should the behavior be verified? Round-trip representative and malformed records, including boundary values.	**Comment vérifier le comportement ?** Tester aller-retour, données mal formées et valeurs limites.

ENGINEERING NOTE / NOTE TECHNIQUE

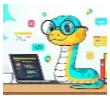
ENGINEERING NOTE / NOTE TECHNIQUE: UUIDs identify tests or acquisitions without central coordination, but they do not prove integrity, ordering or authenticity. / Les UUID identifient les essais sans coordination centrale, mais ne prouvent ni intégrité, ni ordre, ni authenticité.

CODE / API EXAMPLE

```
from uuid import uuid4
run_id=str(uuid4())
print(len(run_id),run_id.count('-'))
```



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Chapter 224 - Logging Fundamentals

ENGLISH	FRANÇAIS
What does Logging Fundamentals actually do? logging provides severity-filtered, structured diagnostics; configure handlers once and avoid duplicate emission or secrets.	**Quel est le rôle de Logging Fundamentals ?** logging fournit des diagnostics par gravité; configurer les handlers sans doublons et ne jamais journaliser de secrets.
What is the engineering design constraint? Make diagnostics attributable and bounded without disclosing credentials or invalid measurements.	**Quelle contrainte de conception faut-il respecter ?** Garder des diagnostics traçables et bornés sans exposer secrets ou mesures invalides.
How should the behavior be verified? Verify filtering, rotation, correlation fields and recovery after interruption.	**Comment vérifier le comportement ?** Vérifier filtres, rotation, traçabilité et reprise après interruption.

ENGINEERING NOTE / NOTE TECHNIQUE

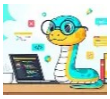
ENGINEERING NOTE / NOTE TECHNIQUE: logging provides severity-filtered, structured diagnostics; configure handlers once and avoid duplicate emission or secrets. / logging fournit des diagnostics par gravité; configurer les handlers sans doublons et ne jamais journaliser de secrets.

CODE / API EXAMPLE

```
import logging
log=logging.getLogger('ahds')
log.addHandler(logging.NullHandler())
log.info('acquisition ready')
```



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Chapter 225 - Logging Levels

ENGLISH	FRANÇAIS
What does Logging Levels actually do? Choose DEBUG for traces, INFO for normal milestones, WARNING for recoverable degradation and ERROR/CRITICAL for failures with operational impact.	**Quel est le rôle de Logging Levels ?** DEBUG trace, INFO activité normale, WARNING dégradation, ERROR/CRITICAL panne et impact important.
What is the engineering design constraint? Make diagnostics attributable and bounded without disclosing credentials or invalid measurements.	**Quelle contrainte de conception faut-il respecter ?** Garder des diagnostics traçables et bornés sans exposer secrets ou mesures invalides.
How should the behavior be verified? Verify filtering, rotation, correlation fields and recovery after interruption.	**Comment vérifier le comportement ?** Vérifier filtres, rotation, traçabilité et reprise après interruption.

ENGINEERING NOTE / NOTE TECHNIQUE

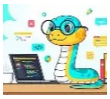
ENGINEERING NOTE / NOTE TECHNIQUE: Choose DEBUG for traces, INFO for normal milestones, WARNING for recoverable degradation and ERROR/CRITICAL for failures with operational impact. / DEBUG trace, INFO activité normale, WARNING dégradation, ERROR/CRITICAL panne et impact important.

CODE / API EXAMPLE

```
import logging
log=logging.getLogger('daq'); log.setLevel(logging.WARNING)
log.warning('calibration approaching expiry')
```



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Chapter 226 - Log Formatting

ENGLISH	FRANÇAIS
What does Log Formatting actually do? Format logs with timestamps, process/thread and correlation IDs; use timezone-aware UTC when events cross machines.	**Quel est le rôle de Log Formatting ?** Formater les journaux avec heure, processus et identifiant de corrélation; privilégier UTC entre machines.
What is the engineering design constraint? Make diagnostics attributable and bounded without disclosing credentials or invalid measurements.	**Quelle contrainte de conception faut-il respecter ?** Garder des diagnostics traçables et bornés sans exposer secrets ou mesures invalides.
How should the behavior be verified? Verify filtering, rotation, correlation fields and recovery after interruption.	**Comment vérifier le comportement ?** Vérifier filtres, rotation, traçabilité et reprise après interruption.

ENGINEERING NOTE / NOTE TECHNIQUE

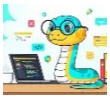
ENGINEERING NOTE / NOTE TECHNIQUE: Format logs with timestamps, process/thread and correlation IDs; use timezone-aware UTC when events cross machines. / Formater les journaux avec heure, processus et identifiant de corrélation; privilégier UTC entre machines.

CODE / API EXAMPLE

```
import logging
handler=logging.StreamHandler()
handler.setFormatter(logging.Formatter('%(levelname)s %(message)s'))
logger=logging.getLogger('testlog');logger.addHandler(handler)
logger.warning('link degraded')
```



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Chapter 227 - Rotating Logs

ENGLISH	FRANÇAIS
What does Rotating Logs actually do? RotatingFileHandler limits file size or time window; rotation is not a substitute for disk quotas, archival policy or crash durability.	**Quel est le rôle de Rotating Logs ?** RotatingFileHandler limite la taille ou durée; ajouter quotas, archivage et stratégie après crash.
What is the engineering design constraint? Make diagnostics attributable and bounded without disclosing credentials or invalid measurements.	**Quelle contrainte de conception faut-il respecter ?** Garder des diagnostics traçables et bornés sans exposer secrets ou mesures invalides.
How should the behavior be verified? Verify filtering, rotation, correlation fields and recovery after interruption.	**Comment vérifier le comportement ?** Vérifier filtres, rotation, traçabilité et reprise après interruption.

ENGINEERING NOTE / NOTE TECHNIQUE

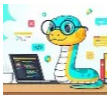
ENGINEERING NOTE / NOTE TECHNIQUE: RotatingFileHandler limits file size or time window; rotation is not a substitute for disk quotas, archival policy or crash durability. / RotatingFileHandler limite la taille ou durée; ajouter quotas, archivage et stratégie après crash.

CODE / API EXAMPLE

```
from logging.handlers import RotatingFileHandler
h=RotatingFileHandler('engineering.log',maxBytes=100000,backupCount=3)
h.close()
```



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Chapter 228 - Logging Handlers

ENGLISH	FRANÇAIS
What does Logging Handlers actually do? Handlers route log records to streams, files, sockets or queues; avoid blocking acquisition threads on slow destinations.	**Quel est le rôle de Logging Handlers ?** Les handlers dirigent les messages vers fichiers, flux ou files; éviter de bloquer l'acquisition sur une destination lente.
What is the engineering design constraint? Make diagnostics attributable and bounded without disclosing credentials or invalid measurements.	**Quelle contrainte de conception faut-il respecter ?** Garder des diagnostics traçables et bornés sans exposer secrets ou mesures invalides.
How should the behavior be verified? Verify filtering, rotation, correlation fields and recovery after interruption.	**Comment vérifier le comportement ?** Vérifier filtres, rotation, traçabilité et reprise après interruption.

ENGINEERING NOTE / NOTE TECHNIQUE

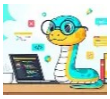
ENGINEERING NOTE / NOTE TECHNIQUE: Handlers route log records to streams, files, sockets or queues; avoid blocking acquisition threads on slow destinations. / Les handlers dirigent les messages vers fichiers, flux ou files; éviter de bloquer l'acquisition sur une destination lente.

CODE / API EXAMPLE

```
import logging
logger=logging.getLogger('instrument')
logger.addHandler(logging.NullHandler())
logger.info('device connected')
```



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Chapter 229 - Structured Logging

ENGLISH	FRANÇAIS
What does Structured Logging actually do? Structured logging stores stable field names, not only prose; redact secrets and define schema evolution for monitoring tools.	**Quel est le rôle de Structured Logging ?** Un journal structuré garde des champs stables; retirer les secrets et versionner le schéma exploité.
What is the engineering design constraint? Make diagnostics attributable and bounded without disclosing credentials or invalid measurements.	**Quelle contrainte de conception faut-il respecter ?** Garder des diagnostics traçables et bornés sans exposer secrets ou mesures invalides.
How should the behavior be verified? Verify filtering, rotation, correlation fields and recovery after interruption.	**Comment vérifier le comportement ?** Vérifier filtres, rotation, traçabilité et reprise après interruption.

ENGINEERING NOTE / NOTE TECHNIQUE

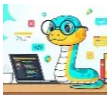
ENGINEERING NOTE / NOTE TECHNIQUE: Structured logging stores stable field names, not only prose; redact secrets and define schema evolution for monitoring tools. / Un journal structuré garde des champs stables; retirer les secrets et versionner le schéma exploité.

CODE / API EXAMPLE

```
import json
print(json.dumps({"event":"adc_overrange","channel":3,"severity":"warning"}))
```



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Chapter 230 - Hardware & Test-System Logging

ENGLISH	FRANÇAIS
What does Hardware & Test-System Logging actually do? Hardware logs must correlate command, response, firmware revision and time source; record invalid measurement status without stale substituted values.	**Quel est le rôle de Hardware & Test-System Logging ?** Journaliser commande, réponse, firmware et horloge; signaler une mesure invalide sans réutiliser une valeur périmée.
What is the engineering design constraint? Make diagnostics attributable and bounded without disclosing credentials or invalid measurements.	**Quelle contrainte de conception faut-il respecter ?** Garder des diagnostics traçables et bornés sans exposer secrets ou mesures invalides.
How should the behavior be verified? Verify filtering, rotation, correlation fields and recovery after interruption.	**Comment vérifier le comportement ?** Vérifier filtres, rotation, traçabilité et reprise après interruption.

ENGINEERING NOTE / NOTE TECHNIQUE

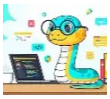
ENGINEERING NOTE / NOTE TECHNIQUE: Hardware logs must correlate command, response, firmware revision and time source; record invalid measurement status without stale substituted values. / Journaliser commande, réponse, firmware et horloge; signaler une mesure invalide sans réutiliser une valeur périmée.

CODE / API EXAMPLE

```
event={"cmd_id":12,"channel":2,"fw":"1.4","status":"timeout"}
print(event["status"])
```



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Chapter 231 - Configuration Management

ENGLISH	FRANÇAIS
What does Configuration Management actually do? Keep default configuration, validated overrides and runtime state separate; store provenance and revision of each applied setting.	**Quel est le rôle de Configuration Management ?** Séparer valeurs par défaut, surcharges validées et état courant; tracer origine et révision de chaque réglage.
What is the engineering design constraint? Make diagnostics attributable and bounded without disclosing credentials or invalid measurements.	**Quelle contrainte de conception faut-il respecter ?** Garder des diagnostics traçables et bornés sans exposer secrets ou mesures invalides.
How should the behavior be verified? Verify filtering, rotation, correlation fields and recovery after interruption.	**Comment vérifier le comportement ?** Vérifier filtres, rotation, traçabilité et reprise après interruption.

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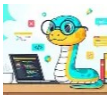
ENGINEERING NOTE / NOTE TECHNIQUE: Keep default configuration, validated overrides and runtime state separate; store provenance and revision of each applied setting. / Séparer valeurs par défaut, surcharges validées et état courant; tracer origine et révision de chaque réglage.

CODE / API EXAMPLE

```
defaults={"timeout_s":1.0}
overrides={"timeout_s":2.0}
settings={**defaults,**overrides}
print(settings)
```



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Chapter 232 - Runtime Configuration

ENGLISH	FRANÇAIS
What does Runtime Configuration actually do? Runtime changes must be validated and applied atomically at a safe boundary; never reconfigure active equipment halfway through a frame.	**Quel est le rôle de Runtime Configuration ?** Valider les changements à chaud et les appliquer à une frontière sûre, pas au milieu d'une mesure.
What is the engineering design constraint? Make diagnostics attributable and bounded without disclosing credentials or invalid measurements.	**Quelle contrainte de conception faut-il respecter ?** Garder des diagnostics traçables et bornés sans exposer secrets ou mesures invalides.
How should the behavior be verified? Verify filtering, rotation, correlation fields and recovery after interruption.	**Comment vérifier le comportement ?** Vérifier filtres, rotation, traçabilité et reprise après interruption.

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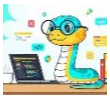
ENGINEERING NOTE / NOTE TECHNIQUE: Runtime changes must be validated and applied atomically at a safe boundary; never reconfigure active equipment halfway through a frame. / Valider les changements à chaud et les appliquer à une frontière sûre, pas au milieu d'une mesure.

CODE / API EXAMPLE

```
current={"rate_hz":1000}
proposed={"rate_hz":2000}
if proposed["rate_hz"] not in (1000,2000): raise ValueError
current=proposed.copy()
```



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Chapter 233 - Persistent Application State

ENGLISH	FRANÇAIS
What does Persistent Application State actually do? Persist only the state needed for recovery, not live handles or unsafe actuator state; version and validate restored records.	**Quel est le rôle de Persistent Application State ?** Conserver les seules données nécessaires au redémarrage; ne pas restaurer aveuglément des handles ou sorties actives.
What is the engineering design constraint? Make diagnostics attributable and bounded without disclosing credentials or invalid measurements.	**Quelle contrainte de conception faut-il respecter ?** Garder des diagnostics traçables et bornés sans exposer secrets ou mesures invalides.
How should the behavior be verified? Verify filtering, rotation, correlation fields and recovery after interruption.	**Comment vérifier le comportement ?** Vérifier filtres, rotation, traçabilité et reprise après interruption.

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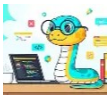
ENGINEERING NOTE / NOTE TECHNIQUE: Persist only the state needed for recovery, not live handles or unsafe actuator state; version and validate restored records. / Conserver les seules données nécessaires au redémarrage; ne pas restaurer aveuglément des handles ou sorties actives.

CODE / API EXAMPLE

```
state={"schema":1,"last_seq":42,"outputs_enabled":False}
assert state["outputs_enabled"] is False
```



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Chapter 234 - Saving & Restoring State

ENGLISH	FRANÇAIS
What does Saving & Restoring State actually do? Saving and restoring state requires schema validation, atomic replacement and clear behavior for missing or corrupt snapshots.	**Quel est le rôle de Saving & Restoring State ?** Sauvegarde et restauration exigent validation, remplacement atomique et stratégie si le fichier manque ou est corrompu.
What is the engineering design constraint? Make diagnostics attributable and bounded without disclosing credentials or invalid measurements.	**Quelle contrainte de conception faut-il respecter ?** Garder des diagnostics traçables et bornés sans exposer secrets ou mesures invalides.
How should the behavior be verified? Verify filtering, rotation, correlation fields and recovery after interruption.	**Comment vérifier le comportement ?** Vérifier filtres, rotation, traçabilité et reprise après interruption.

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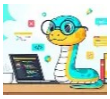
ENGINEERING NOTE / NOTE TECHNIQUE: Saving and restoring state requires schema validation, atomic replacement and clear behavior for missing or corrupt snapshots. / Sauvegarde et restauration exigent validation, remplacement atomique et stratégie si le fichier manque ou est corrompu.

CODE / API EXAMPLE

```
import json
blob=json.dumps({"version":1,"seq":17})
snapshot=json.loads(blob)
if snapshot.get("version")!=1: raise ValueError("schema")
```



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Chapter 235 - Auditability & Traceability

ENGLISH	FRANÇAIS
What does Auditability & Traceability actually do? Audit trails must attribute events to time, actor, software/config version and outcome; append-only intent still needs access control and integrity measures.	**Quel est le rôle de Auditability & Traceability ?** La traçabilité associe acteur, heure, version et résultat; un journal d'audit exige aussi contrôle d'accès et intégrité.
What is the engineering design constraint? Make diagnostics attributable and bounded without disclosing credentials or invalid measurements.	**Quelle contrainte de conception faut-il respecter ?** Garder des diagnostics traçables et bornés sans exposer secrets ou mesures invalides.
How should the behavior be verified? Verify filtering, rotation, correlation fields and recovery after interruption.	**Comment vérifier le comportement ?** Vérifier filtres, rotation, traçabilité et reprise après interruption.

ENGINEERING NOTE / NOTE TECHNIQUE

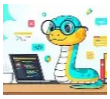
ENGINEERING NOTE / NOTE TECHNIQUE: Audit trails must attribute events to time, actor, software/config version and outcome; append-only intent still needs access control and integrity measures. / La traçabilité associe acteur, heure, version et résultat; un journal d'audit exige aussi contrôle d'accès et intégrité.

CODE / API EXAMPLE

```
audit={"actor":"service","action":"set_rate","result":"ok","revision":"C"}
print(sorted(audit))
```



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Chapter 236 - Reliable Python Applications

ENGLISH	FRANÇAIS
What does Reliable Python Applications actually do? Reliability comes from explicit failure modes, bounded operations, instrumentation and tested recovery, not merely adding exception handlers.	**Quel est le rôle de Reliable Python Applications ?** La fiabilité repose sur modes de panne, opérations bornées, instrumentation et reprise testée, pas seulement except.
What is the engineering design constraint? Define safe degraded states, bounded waits and recovery ownership before deployment.	**Quelle contrainte de conception faut-il respecter ?** Définir états dégradés sûrs, attentes bornées et responsabilité de reprise.
How should the behavior be verified? Inject timeout, disconnect, power loss and repeated failure, then validate state.	**Comment vérifier le comportement ?** Injecter délai, déconnexion, coupure et panne répétée; vérifier l'état.

ENGINEERING NOTE / NOTE TECHNIQUE

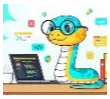
ENGINEERING NOTE / NOTE TECHNIQUE: Reliability comes from explicit failure modes, bounded operations, instrumentation and tested recovery, not merely adding exception handlers. / La fiabilité repose sur modes de panne, opérations bornées, instrumentation et reprise testée, pas seulement except.

CODE / API EXAMPLE

```
def stable_average(values):
   if not values: raise ValueError("no samples")
   return sum(values)/len(values)
print(stable_average([1,2,3]))
```



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Chapter 237 - Fault-Tolerant Design Principles

ENGLISH	FRANÇAIS
What does Fault-Tolerant Design Principles actually do? A fault-tolerant design defines detection, containment, degraded modes and safe shutdown; some failures must stop the system rather than retry.	**Quel est le rôle de Fault-Tolerant Design Principles ?** Tolérance aux pannes: détection, confinement, mode dégradé et arrêt sûr; certaines erreurs exigent un arrêt, pas un retry.
What is the engineering design constraint? Define safe degraded states, bounded waits and recovery ownership before deployment.	**Quelle contrainte de conception faut-il respecter ?** Définir états dégradés sûrs, attentes bornées et responsabilité de reprise.
How should the behavior be verified? Inject timeout, disconnect, power loss and repeated failure, then validate state.	**Comment vérifier le comportement ?** Injecter délai, déconnexion, coupure et panne répétée; vérifier l'état.

ENGINEERING NOTE / NOTE TECHNIQUE

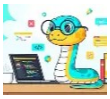
ENGINEERING NOTE / NOTE TECHNIQUE: A fault-tolerant design defines detection, containment, degraded modes and safe shutdown; some failures must stop the system rather than retry. / Tolérance aux pannes: détection, confinement, mode dégradé et arrêt sûr; certaines erreurs exigent un arrêt, pas un retry.

CODE / API EXAMPLE

```
def on_fault(is_safe):
    return "continue" if is_safe else "disable_outputs"
print(on_fault(False))
```



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Chapter 238 - Retry Strategies

ENGLISH	FRANÇAIS
What does Retry Strategies actually do? Retry only idempotent or deduplicated operations after transient faults; cap attempts and honor end-to-end deadlines.	**Quel est le rôle de Retry Strategies ?** Ne réessayer que les opérations idempotentes ou dédupliquées; limiter tentatives et délai global.
What is the engineering design constraint? Define safe degraded states, bounded waits and recovery ownership before deployment.	**Quelle contrainte de conception faut-il respecter ?** Définir états dégradés sûrs, attentes bornées et responsabilité de reprise.
How should the behavior be verified? Inject timeout, disconnect, power loss and repeated failure, then validate state.	**Comment vérifier le comportement ?** Injecter délai, déconnexion, coupure et panne répétée; vérifier l'état.

ENGINEERING NOTE / NOTE TECHNIQUE

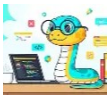
ENGINEERING NOTE / NOTE TECHNIQUE: Retry only idempotent or deduplicated operations after transient faults; cap attempts and honor end-to-end deadlines. / Ne réessayer que les opérations idempotentes ou dédupliquées; limiter tentatives et délai global.

CODE / API EXAMPLE

```
attempts=0
for attempt in range(3):
   attempts+=1
   ok=(attempt==2)
   if ok: break
else: raise TimeoutError("attempts exhausted")
print(attempts)
```



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Chapter 239 - Timeouts

ENGLISH	FRANÇAIS
What does Timeouts actually do? Timeouts bound blocking operations but require explicit scopes and deadlines; a timeout does not guarantee cancellation of a stuck native driver.	**Quel est le rôle de Timeouts ?** Les timeouts bornent l'attente avec une échéance définie; ils ne peuvent pas forcément annuler un pilote natif bloqué.
What is the engineering design constraint? Define safe degraded states, bounded waits and recovery ownership before deployment.	**Quelle contrainte de conception faut-il respecter ?** Définir états dégradés sûrs, attentes bornées et responsabilité de reprise.
How should the behavior be verified? Inject timeout, disconnect, power loss and repeated failure, then validate state.	**Comment vérifier le comportement ?** Injecter délai, déconnexion, coupure et panne répétée; vérifier l'état.

ENGINEERING NOTE / NOTE TECHNIQUE

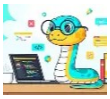
ENGINEERING NOTE / NOTE TECHNIQUE: Timeouts bound blocking operations but require explicit scopes and deadlines; a timeout does not guarantee cancellation of a stuck native driver. / Les timeouts bornent l'attente avec une échéance définie; ils ne peuvent pas forcément annuler un pilote natif bloqué.

CODE / API EXAMPLE

```
import time
deadline=time.monotonic()+0.1
while time.monotonic()<deadline:
    break
else: raise TimeoutError("deadline reached")
```



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Chapter 240 - Backoff Strategies

ENGLISH	FRANÇAIS
What does Backoff Strategies actually do? Exponential backoff with jitter reduces synchronized reconnection storms; enforce a maximum delay and a total retry budget.	**Quel est le rôle de Backoff Strategies ?** Le backoff exponentiel avec jitter limite les reconnexions simultanées; fixer délai maximal et budget total.
What is the engineering design constraint? Define safe degraded states, bounded waits and recovery ownership before deployment.	**Quelle contrainte de conception faut-il respecter ?** Définir états dégradés sûrs, attentes bornées et responsabilité de reprise.
How should the behavior be verified? Inject timeout, disconnect, power loss and repeated failure, then validate state.	**Comment vérifier le comportement ?** Injecter délai, déconnexion, coupure et panne répétée; vérifier l'état.

ENGINEERING NOTE / NOTE TECHNIQUE

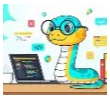
ENGINEERING NOTE / NOTE TECHNIQUE: Exponential backoff with jitter reduces synchronized reconnection storms; enforce a maximum delay and a total retry budget. / Le backoff exponentiel avec jitter limite les reconnexions simultanées; fixer délai maximal et budget total.

CODE / API EXAMPLE

```
import random
for attempt in range(3):
  delay=min(5.0,0.2*2**attempt)*random.uniform(0.8,1.2)
  print(round(delay,3))
```



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Chapter 241 - Software & Hardware Watchdog Concepts

ENGLISH	FRANÇAIS
What does Software & Hardware Watchdog Concepts actually do? Watchdogs detect absent progress, not correctness; refresh only after verified work and set recovery to a known safe hardware state.	**Quel est le rôle de Software & Hardware Watchdog Concepts ?** Un watchdog détecte l'absence de progression, non l'exactitude; ne le rafraîchir qu'après une opération validée.
What is the engineering design constraint? Define safe degraded states, bounded waits and recovery ownership before deployment.	**Quelle contrainte de conception faut-il respecter ?** Définir états dégradés sûrs, attentes bornées et responsabilité de reprise.
How should the behavior be verified? Inject timeout, disconnect, power loss and repeated failure, then validate state.	**Comment vérifier le comportement ?** Injecter délai, déconnexion, coupure et panne répétée; vérifier l'état.

ENGINEERING NOTE / NOTE TECHNIQUE

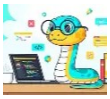
ENGINEERING NOTE / NOTE TECHNIQUE: Watchdogs detect absent progress, not correctness; refresh only after verified work and set recovery to a known safe hardware state. / Un watchdog détecte l'absence de progression, non l'exactitude; ne le rafraîchir qu'après une opération validée.

CODE / API EXAMPLE

```
progress={"last_heartbeat":10,"now":16}
expired=progress["now"]-progress["last_heartbeat"]>5
print("restart" if expired else "healthy")
```



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Chapter 242 - Process Monitoring

ENGLISH	FRANÇAIS
What does Process Monitoring actually do? Process supervisors track liveness and restart budgets; crash loops must be contained and logged rather than restarted indefinitely.	**Quel est le rôle de Process Monitoring ?** Un superviseur suit la disponibilité et limite les redémarrages; empêcher les boucles de crash répétées.
What is the engineering design constraint? Define safe degraded states, bounded waits and recovery ownership before deployment.	**Quelle contrainte de conception faut-il respecter ?** Définir états dégradés sûrs, attentes bornées et responsabilité de reprise.
How should the behavior be verified? Inject timeout, disconnect, power loss and repeated failure, then validate state.	**Comment vérifier le comportement ?** Injecter délai, déconnexion, coupure et panne répétée; vérifier l'état.

ENGINEERING NOTE / NOTE TECHNIQUE

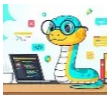
ENGINEERING NOTE / NOTE TECHNIQUE: Process supervisors track liveness and restart budgets; crash loops must be contained and logged rather than restarted indefinitely. / Un superviseur suit la disponibilité et limite les redémarrages; empêcher les boucles de crash répétées.

CODE / API EXAMPLE

```
restarts=4; budget=3
if restarts>budget:
    print("escalate instead of restart")
```



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Chapter 243 - Disconnect & Reconnect Strategies

ENGLISH	FRANÇAIS
What does Disconnect & Reconnect Strategies actually do? Reconnect logic must invalidate old sessions and pending requests; negotiate identity/configuration again before accepting new readings.	**Quel est le rôle de Disconnect & Reconnect Strategies ?** Après reconnexion, invalider l'ancienne session et les commandes en attente; revérifier identité et configuration.
What is the engineering design constraint? Define safe degraded states, bounded waits and recovery ownership before deployment.	**Quelle contrainte de conception faut-il respecter ?** Définir états dégradés sûrs, attentes bornées et responsabilité de reprise.
How should the behavior be verified? Inject timeout, disconnect, power loss and repeated failure, then validate state.	**Comment vérifier le comportement ?** Injecter délai, déconnexion, coupure et panne répétée; vérifier l'état.

ENGINEERING NOTE / NOTE TECHNIQUE

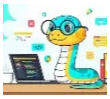
ENGINEERING NOTE / NOTE TECHNIQUE: Reconnect logic must invalidate old sessions and pending requests; negotiate identity/configuration again before accepting new readings. / Après reconnexion, invalider l'ancienne session et les commandes en attente; revérifier identité et configuration.

CODE / API EXAMPLE

```
state="disconnected"
if state=="disconnected": state="connecting"
if state=="connecting": state="verify_device"
print(state)
```



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Chapter 244 - Communication Recovery

ENGLISH	FRANÇAIS
What does Communication Recovery actually do? Communication recovery addresses sequence gaps, malformed frames and partial commands; distinguish safe retransmission from commands with side effects.	**Quel est le rôle de Communication Recovery ?** Récupérer une liaison implique séquences, trames corrompues et commandes partielles; attention aux actions non idempotentes.
What is the engineering design constraint? Define safe degraded states, bounded waits and recovery ownership before deployment.	**Quelle contrainte de conception faut-il respecter ?** Définir états dégradés sûrs, attentes bornées et responsabilité de reprise.
How should the behavior be verified? Inject timeout, disconnect, power loss and repeated failure, then validate state.	**Comment vérifier le comportement ?** Injecter délai, déconnexion, coupure et panne répétée; vérifier l'état.

ENGINEERING NOTE / NOTE TECHNIQUE

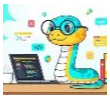
ENGINEERING NOTE / NOTE TECHNIQUE: Communication recovery addresses sequence gaps, malformed frames and partial commands; distinguish safe retransmission from commands with side effects. / Récupérer une liaison implique séquences, trames corrompues et commandes partielles; attention aux actions non idempotentes.

CODE / API EXAMPLE

```
last_seq=40; received_seq=42
if received_seq!=last_seq+1:
    print("gap detected; request resync")
```



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Chapter 245 - Data Integrity During Failure

ENGLISH	FRANÇAIS
What does Data Integrity During Failure actually do? Protect data integrity against interruption with checksums, sequence numbers, atomic commits and explicit validity markers.	**Quel est le rôle de Data Integrity During Failure ?** Protéger l'intégrité avec checksum, séquence, validation et commit atomique; rendre les mesures incomplètes invalides.
What is the engineering design constraint? Define safe degraded states, bounded waits and recovery ownership before deployment.	**Quelle contrainte de conception faut-il respecter ?** Définir états dégradés sûrs, attentes bornées et responsabilité de reprise.
How should the behavior be verified? Inject timeout, disconnect, power loss and repeated failure, then validate state.	**Comment vérifier le comportement ?** Injecter délai, déconnexion, coupure et panne répétée; vérifier l'état.

ENGINEERING NOTE / NOTE TECHNIQUE

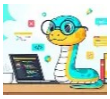
ENGINEERING NOTE / NOTE TECHNIQUE: Protect data integrity against interruption with checksums, sequence numbers, atomic commits and explicit validity markers. / Protéger l'intégrité avec checksum, séquence, validation et commit atomique; rendre les mesures incomplètes invalides.

CODE / API EXAMPLE

```
import hashlib
frame=b"sensor:17"
digest=hashlib.sha256(frame).hexdigest()
print(len(digest))
```



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Chapter 246 - Crash-Safe Logging

ENGLISH	FRANÇAIS
What does Crash-Safe Logging actually do? Crash-safe logging requires explicit flushing/durability tradeoffs; an application log handler does not automatically guarantee fsync on every event.	**Quel est le rôle de Crash-Safe Logging ?** La journalisation résistante aux crashs exige un choix explicite entre débit et persistance; flush n'est pas toujours fsync.
What is the engineering design constraint? Define safe degraded states, bounded waits and recovery ownership before deployment.	**Quelle contrainte de conception faut-il respecter ?** Définir états dégradés sûrs, attentes bornées et responsabilité de reprise.
How should the behavior be verified? Inject timeout, disconnect, power loss and repeated failure, then validate state.	**Comment vérifier le comportement ?** Injecter délai, déconnexion, coupure et panne répétée; vérifier l'état.

ENGINEERING NOTE / NOTE TECHNIQUE

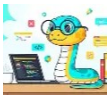
ENGINEERING NOTE / NOTE TECHNIQUE: Crash-safe logging requires explicit flushing/durability tradeoffs; an application log handler does not automatically guarantee fsync on every event. / La journalisation résistante aux crashs exige un choix explicite entre débit et persistance; flush n'est pas toujours fsync.

CODE / API EXAMPLE

```
import os,tempfile
with tempfile.TemporaryFile() as f:
    f.write(b"fault\n"); f.flush()
    os.fsync(f.fileno())
```



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Chapter 247 - Configuration Recovery

ENGLISH	FRANÇAIS
What does Configuration Recovery actually do? Recover configuration from a validated last-known-good snapshot; never silently apply corrupt or unknown settings to hardware.	**Quel est le rôle de Configuration Recovery ?** Restaurer une configuration connue et validée; ne jamais appliquer un fichier corrompu ou de version inconnue.
What is the engineering design constraint? Define safe degraded states, bounded waits and recovery ownership before deployment.	**Quelle contrainte de conception faut-il respecter ?** Définir états dégradés sûrs, attentes bornées et responsabilité de reprise.
How should the behavior be verified? Inject timeout, disconnect, power loss and repeated failure, then validate state.	**Comment vérifier le comportement ?** Injecter délai, déconnexion, coupure et panne répétée; vérifier l'état.

ENGINEERING NOTE / NOTE TECHNIQUE

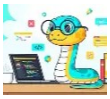
ENGINEERING NOTE / NOTE TECHNIQUE: Recover configuration from a validated last-known-good snapshot; never silently apply corrupt or unknown settings to hardware. / Restaurer une configuration connue et validée; ne jamais appliquer un fichier corrompu ou de version inconnue.

CODE / API EXAMPLE

```
snapshot={"schema":1,"safe_mode":True}
if snapshot.get("schema")!=1: raise ValueError("schema")
print(snapshot["safe_mode"])
```



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Chapter 248 - Resource-Leak Awareness

ENGLISH	FRANÇAIS
What does Resource-Leak Awareness actually do? Track file descriptors, threads, buffers and native allocations in long-running services; repeated reconnects often reveal leaks.	**Quel est le rôle de Resource-Leak Awareness ?** Surveiller handles, threads, tampons et allocations dans un service permanent; tester les cycles de reconnexion.
What is the engineering design constraint? Define safe degraded states, bounded waits and recovery ownership before deployment.	**Quelle contrainte de conception faut-il respecter ?** Définir états dégradés sûrs, attentes bornées et responsabilité de reprise.
How should the behavior be verified? Inject timeout, disconnect, power loss and repeated failure, then validate state.	**Comment vérifier le comportement ?** Injecter délai, déconnexion, coupure et panne répétée; vérifier l'état.

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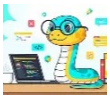
ENGINEERING NOTE / NOTE TECHNIQUE: Track file descriptors, threads, buffers and native allocations in long-running services; repeated reconnects often reveal leaks. / Surveiller handles, threads, tampons et allocations dans un service permanent; tester les cycles de reconnexion.

CODE / API EXAMPLE

```
from contextlib import closing
from io import BytesIO
for _ in range(100):
    with closing(BytesIO(b"ok")) as f: assert f.read()==b"ok"
```



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Chapter 249 - Graceful Shutdown

ENGLISH	FRANÇAIS
What does Graceful Shutdown actually do? A graceful shutdown stops incoming work, drives outputs safe, drains bounded queues, flushes state and releases device handles in order.	**Quel est le rôle de Graceful Shutdown ?** Un arrêt propre cesse les nouvelles commandes, sécurise les sorties, vide les files bornées et libère les ressources.
What is the engineering design constraint? Define safe degraded states, bounded waits and recovery ownership before deployment.	**Quelle contrainte de conception faut-il respecter ?** Définir états dégradés sûrs, attentes bornées et responsabilité de reprise.
How should the behavior be verified? Inject timeout, disconnect, power loss and repeated failure, then validate state.	**Comment vérifier le comportement ?** Injecter délai, déconnexion, coupure et panne répétée; vérifier l'état.

ENGINEERING NOTE / NOTE TECHNIQUE

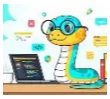
ENGINEERING NOTE / NOTE TECHNIQUE: A graceful shutdown stops incoming work, drives outputs safe, drains bounded queues, flushes state and releases device handles in order. / Un arrêt propre cesse les nouvelles commandes, sécurise les sorties, vide les files bornées et libère les ressources.

CODE / API EXAMPLE

```
steps=["stop commands","safe outputs","flush records","close ports"]
for step in steps: print(step)
```



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Chapter 250 - Power-Loss Recovery

ENGLISH	FRANÇAIS
What does Power-Loss Recovery actually do? Power loss interrupts arbitrary operations; design durable checkpoints, brownout-safe hardware behavior and restart validation.	**Quel est le rôle de Power-Loss Recovery ?** Une coupure peut interrompre toute opération; prévoir points de reprise durables et état matériel sûr au redémarrage.
What is the engineering design constraint? Define safe degraded states, bounded waits and recovery ownership before deployment.	**Quelle contrainte de conception faut-il respecter ?** Définir états dégradés sûrs, attentes bornées et responsabilité de reprise.
How should the behavior be verified? Inject timeout, disconnect, power loss and repeated failure, then validate state.	**Comment vérifier le comportement ?** Injecter délai, déconnexion, coupure et panne répétée; vérifier l'état.

ENGINEERING NOTE / NOTE TECHNIQUE

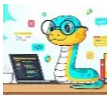
ENGINEERING NOTE / NOTE TECHNIQUE: Power loss interrupts arbitrary operations; design durable checkpoints, brownout-safe hardware behavior and restart validation. / Une coupure peut interrompre toute opération; prévoir points de reprise durables et état matériel sûr au redémarrage.

CODE / API EXAMPLE

```
checkpoint={"schema":1,"last_committed_seq":128}
resume_from=checkpoint["last_committed_seq"]+1
print(resume_from)
```



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Chapter 251 - Restart & Resume Strategies

ENGLISH	FRANÇAIS
What does Restart & Resume Strategies actually do? Resume only from committed progress; idempotency keys or deduplication prevent duplicate actuator commands and duplicate measurement records.	**Quel est le rôle de Restart & Resume Strategies ?** Reprendre après le dernier commit; utiliser clé d'idempotence ou déduplication pour éviter des actions répétées.
What is the engineering design constraint? Define safe degraded states, bounded waits and recovery ownership before deployment.	**Quelle contrainte de conception faut-il respecter ?** Définir états dégradés sûrs, attentes bornées et responsabilité de reprise.
How should the behavior be verified? Inject timeout, disconnect, power loss and repeated failure, then validate state.	**Comment vérifier le comportement ?** Injecter délai, déconnexion, coupure et panne répétée; vérifier l'état.

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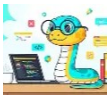
ENGINEERING NOTE / NOTE TECHNIQUE: Resume only from committed progress; idempotency keys or deduplication prevent duplicate actuator commands and duplicate measurement records. / Reprendre après le dernier commit; utiliser clé d'idempotence ou déduplication pour éviter des actions répétées.

CODE / API EXAMPLE

```
seen={"cmd-17"};incoming="cmd-17"
if incoming in seen: print("ignore duplicate")
```



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Chapter 252 - Long-Running / 24x7 Application Architecture

ENGLISH	FRANÇAIS
What does Long-Running / 24x7 Application Architecture actually do? A 24/7 application needs memory limits, bounded queues, health metrics, restart policy, upgrade plans and fault-injection tests.	**Quel est le rôle de Long-Running / 24x7 Application Architecture ?** Une application 24/7 exige files bornées, limites mémoire, métriques de santé et tests de panne/reprise.
What is the engineering design constraint? Define safe degraded states, bounded waits and recovery ownership before deployment.	**Quelle contrainte de conception faut-il respecter ?** Définir états dégradés sûrs, attentes bornées et responsabilité de reprise.
How should the behavior be verified? Inject timeout, disconnect, power loss and repeated failure, then validate state.	**Comment vérifier le comportement ?** Injecter délai, déconnexion, coupure et panne répétée; vérifier l'état.

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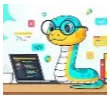
ENGINEERING NOTE / NOTE TECHNIQUE: A 24/7 application needs memory limits, bounded queues, health metrics, restart policy, upgrade plans and fault-injection tests. / Une application 24/7 exige files bornées, limites mémoire, métriques de santé et tests de panne/reprise.

CODE / API EXAMPLE

```
from collections import deque
queue=deque(maxlen=4)
for x in range(7):queue.append(x)
print(list(queue))  # oldest records dropped: document this policy
```



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Chapter 253 - Secure Programming Fundamentals for Engineering Applications

ENGLISH	FRANÇAIS
What does Secure Programming Fundamentals for Engineering Applications actually do? Security starts from assets, attacker capabilities and trust boundaries; use least privilege, defense in depth and validated updates.	**Quel est le rôle de Secure Programming Fundamentals for Engineering Applications ?** La sécurité part des actifs et frontières de confiance; appliquer moindre privilège, défense en profondeur et mises à jour vérifiées.
What is the engineering design constraint? Apply least privilege, validation, explicit trust boundaries and secure failure behavior.	**Quelle contrainte de conception faut-il respecter ?** Appliquer moindre privilège, validation, frontières explicites et échec sûr.
How should the behavior be verified? Test malicious input, unauthorized access and denial-of-service limits.	**Comment vérifier le comportement ?** Tester entrée hostile, accès interdit et limites de ressources.

ENGINEERING NOTE / NOTE TECHNIQUE

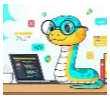
ENGINEERING NOTE / NOTE TECHNIQUE: Security starts from assets, attacker capabilities and trust boundaries; use least privilege, defense in depth and validated updates. / La sécurité part des actifs et frontières de confiance; appliquer moindre privilège, défense en profondeur et mises à jour vérifiées.

CODE / API EXAMPLE

```
def permitted(role, action):
    return role=="engineer" and action=="calibrate"
print(permitted("guest","calibrate"))
```



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Chapter 254 - Safe Input Handling

ENGLISH	FRANÇAIS
What does Safe Input Handling actually do? Validate type, length, encoding, units and allowed ranges at the boundary; never treat input parsing as authorization.	**Quel est le rôle de Safe Input Handling ?** Valider type, longueur, encodage, unités et limites; le décodage d'entrée ne remplace pas l'autorisation.
What is the engineering design constraint? Apply least privilege, validation, explicit trust boundaries and secure failure behavior.	**Quelle contrainte de conception faut-il respecter ?** Appliquer moindre privilège, validation, frontières explicites et échec sûr.
How should the behavior be verified? Test malicious input, unauthorized access and denial-of-service limits.	**Comment vérifier le comportement ?** Tester entrée hostile, accès interdit et limites de ressources.

ENGINEERING NOTE / NOTE TECHNIQUE

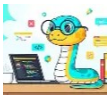
ENGINEERING NOTE / NOTE TECHNIQUE: Validate type, length, encoding, units and allowed ranges at the boundary; never treat input parsing as authorization. / Valider type, longueur, encodage, unités et limites; le décodage d'entrée ne remplace pas l'autorisation.

CODE / API EXAMPLE

```
def parse_channel(text):
    channel=int(text)
    if not 1<=channel<=16: raise ValueError("channel range")
    return channel
print(parse_channel("4"))
```



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Chapter 255 - Safe Path & File Handling

ENGLISH	FRANÇAIS
What does Safe Path & File Handling actually do? Normalize and constrain filesystem paths, accounting for symlink races; check resolved targets and use OS-level confinement for hostile inputs.	**Quel est le rôle de Safe Path & File Handling ?** Normaliser les chemins et limiter leur périmètre; tenir compte des liens symboliques et risques de course.
What is the engineering design constraint? Apply least privilege, validation, explicit trust boundaries and secure failure behavior.	**Quelle contrainte de conception faut-il respecter ?** Appliquer moindre privilège, validation, frontières explicites et échec sûr.
How should the behavior be verified? Test malicious input, unauthorized access and denial-of-service limits.	**Comment vérifier le comportement ?** Tester entrée hostile, accès interdit et limites de ressources.

ENGINEERING NOTE / NOTE TECHNIQUE

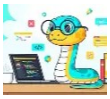
ENGINEERING NOTE / NOTE TECHNIQUE: Normalize and constrain filesystem paths, accounting for symlink races; check resolved targets and use OS-level confinement for hostile inputs. / Normaliser les chemins et limiter leur périmètre; tenir compte des liens symboliques et risques de course.

CODE / API EXAMPLE

```
from pathlib import Path
root=Path("/tmp/measurements").resolve()
requested=(root/"run01.csv").resolve()
if not requested.is_relative_to(root): raise ValueError("path escape")
```



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Chapter 256 - Secrets & Configuration Safety

ENGLISH	FRANÇAIS
What does Secrets & Configuration Safety actually do? Keep secrets out of source, logs and published configs; use environment/secret stores with restricted access and rotation procedures.	**Quel est le rôle de Secrets & Configuration Safety ?** Éviter les secrets dans code, journaux et configs publiées; employer coffre ou environnement protégé avec rotation.
What is the engineering design constraint? Apply least privilege, validation, explicit trust boundaries and secure failure behavior.	**Quelle contrainte de conception faut-il respecter ?** Appliquer moindre privilège, validation, frontières explicites et échec sûr.
How should the behavior be verified? Test malicious input, unauthorized access and denial-of-service limits.	**Comment vérifier le comportement ?** Tester entrée hostile, accès interdit et limites de ressources.

ENGINEERING NOTE / NOTE TECHNIQUE

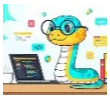
ENGINEERING NOTE / NOTE TECHNIQUE: Keep secrets out of source, logs and published configs; use environment/secret stores with restricted access and rotation procedures. / Éviter les secrets dans code, journaux et configs publiées; employer coffre ou environnement protégé avec rotation.

CODE / API EXAMPLE

```
import os
token=os.environ.get("AHDS_DEVICE_TOKEN")
print("configured" if token else "missing")  # never print secret value
```



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Chapter 257 - Safe Use of eval() & exec()

ENGLISH	FRANÇAIS
What does Safe Use of eval() & exec() actually do? eval and exec run Python code and are unsafe for untrusted input; use explicit parsers, allowlisted operations or ast.literal_eval only for suitable literals.	**Quel est le rôle de Safe Use of eval() & exec() ?** eval et exec exécutent du code et ne conviennent pas aux entrées non fiables; préférer analyseurs et opérations autorisées.
What is the engineering design constraint? Apply least privilege, validation, explicit trust boundaries and secure failure behavior.	**Quelle contrainte de conception faut-il respecter ?** Appliquer moindre privilège, validation, frontières explicites et échec sûr.
How should the behavior be verified? Test malicious input, unauthorized access and denial-of-service limits.	**Comment vérifier le comportement ?** Tester entrée hostile, accès interdit et limites de ressources.

ENGINEERING NOTE / NOTE TECHNIQUE

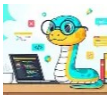
ENGINEERING NOTE / NOTE TECHNIQUE: eval and exec run Python code and are unsafe for untrusted input; use explicit parsers, allowlisted operations or ast.literal_eval only for suitable literals. / eval et exec exécutent du code et ne conviennent pas aux entrées non fiables; préférer analyseurs et opérations autorisées.

CODE / API EXAMPLE

```
import ast
values=ast.literal_eval("[1, 2, 3]")
assert isinstance(values,list) and all(type(v) is int for v in values)
```



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Chapter 258 - Safe Deserialization & pickle Boundaries

ENGLISH	FRANÇAIS
What does Safe Deserialization & pickle Boundaries actually do? Unpickling untrusted bytes is unsafe even when wrapped in try/except; use bounded JSON and schema checks for external messages.	**Quel est le rôle de Safe Deserialization & pickle Boundaries ?** Charger un pickle externe peut exécuter du code, même sous try/except; choisir JSON limité et validé.
What is the engineering design constraint? Apply least privilege, validation, explicit trust boundaries and secure failure behavior.	**Quelle contrainte de conception faut-il respecter ?** Appliquer moindre privilège, validation, frontières explicites et échec sûr.
How should the behavior be verified? Test malicious input, unauthorized access and denial-of-service limits.	**Comment vérifier le comportement ?** Tester entrée hostile, accès interdit et limites de ressources.

ENGINEERING NOTE / NOTE TECHNIQUE

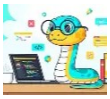
ENGINEERING NOTE / NOTE TECHNIQUE: Unpickling untrusted bytes is unsafe even when wrapped in try/except; use bounded JSON and schema checks for external messages. / Charger un pickle externe peut exécuter du code, même sous try/except; choisir JSON limité et validé.

CODE / API EXAMPLE

```
import json
payload=json.loads('{"gain":1.25}')
if not isinstance(payload.get("gain"),(int,float)):
    raise ValueError("bad payload")
```



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Chapter 259 - Safe subprocess Use

ENGLISH	FRANÇAIS
What does Safe subprocess Use actually do? subprocess should receive a fixed executable and argument list without shell=True for untrusted strings; apply timeouts and inspect return codes.	**Quel est le rôle de Safe subprocess Use ?** Pour subprocess, utiliser une liste d'arguments, sans shell=True pour les données externes; fixer délai et contrôler statut.
What is the engineering design constraint? Apply least privilege, validation, explicit trust boundaries and secure failure behavior.	**Quelle contrainte de conception faut-il respecter ?** Appliquer moindre privilège, validation, frontières explicites et échec sûr.
How should the behavior be verified? Test malicious input, unauthorized access and denial-of-service limits.	**Comment vérifier le comportement ?** Tester entrée hostile, accès interdit et limites de ressources.

ENGINEERING NOTE / NOTE TECHNIQUE

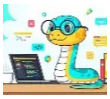
ENGINEERING NOTE / NOTE TECHNIQUE: subprocess should receive a fixed executable and argument list without shell=True for untrusted strings; apply timeouts and inspect return codes. / Pour subprocess, utiliser une liste d'arguments, sans shell=True pour les données externes; fixer délai et contrôler statut.

CODE / API EXAMPLE

```
import subprocess,sys
result=subprocess.run([sys.executable,"-c","print(3)"],capture_output=True,text=True,timeout=5,check=True)
print(result.stdout.strip())
```



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Chapter 260 - Trust Boundaries

ENGLISH	FRANÇAIS
What does Trust Boundaries actually do? Every crossing from user data to files, device commands, network services or privileged code must validate authorization and representation.	**Quel est le rôle de Trust Boundaries ?** Chaque passage vers fichier, commande matérielle, réseau ou privilèges exige validation et autorisation explicites.
What is the engineering design constraint? Apply least privilege, validation, explicit trust boundaries and secure failure behavior.	**Quelle contrainte de conception faut-il respecter ?** Appliquer moindre privilège, validation, frontières explicites et échec sûr.
How should the behavior be verified? Test malicious input, unauthorized access and denial-of-service limits.	**Comment vérifier le comportement ?** Tester entrée hostile, accès interdit et limites de ressources.

ENGINEERING NOTE / NOTE TECHNIQUE

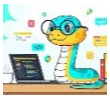
ENGINEERING NOTE / NOTE TECHNIQUE: Every crossing from user data to files, device commands, network services or privileged code must validate authorization and representation. / Chaque passage vers fichier, commande matérielle, réseau ou privilèges exige validation et autorisation explicites.

CODE / API EXAMPLE

```
request={"role":"viewer","command":"write_output"}
if request["role"]!="operator":
    print("deny write at trust boundary")
```



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Chapter 261 - Secure Operational Design Checklist

ENGLISH	FRANÇAIS
What does Secure Operational Design Checklist actually do? Security review includes input boundaries, credentials, dependencies, updates, audit logs, privilege and safe failure; test controls, not just checklist presence.	**Quel est le rôle de Secure Operational Design Checklist ?** La revue sécurité couvre entrées, secrets, dépendances, mises à jour, journaux et repli sûr; tester réellement chaque contrôle.
What is the engineering design constraint? Apply least privilege, validation, explicit trust boundaries and secure failure behavior.	**Quelle contrainte de conception faut-il respecter ?** Appliquer moindre privilège, validation, frontières explicites et échec sûr.
How should the behavior be verified? Test malicious input, unauthorized access and denial-of-service limits.	**Comment vérifier le comportement ?** Tester entrée hostile, accès interdit et limites de ressources.

ENGINEERING NOTE / NOTE TECHNIQUE

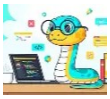
ENGINEERING NOTE / NOTE TECHNIQUE: Security review includes input boundaries, credentials, dependencies, updates, audit logs, privilege and safe failure; test controls, not just checklist presence. / La revue sécurité couvre entrées, secrets, dépendances, mises à jour, journaux et repli sûr; tester réellement chaque contrôle.

CODE / API EXAMPLE

```
checks={"validated_inputs":True,"least_privilege":True,"secrets_redacted":True}
assert all(checks.values())
```



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Chapter 262 - Reliability & Security Integration Checklist

ENGLISH	FRANÇAIS
What does Reliability & Security Integration Checklist actually do? Reliability and security interact: retry limits prevent resource exhaustion, integrity protects recovery state, and fail-safe design handles unknown commands.	**Quel est le rôle de Reliability & Security Integration Checklist ?** Fiabilité et sécurité se rejoignent: retries bornés, état récupéré intègre et refus sûr des commandes inconnues.
What is the engineering design constraint? Apply least privilege, validation, explicit trust boundaries and secure failure behavior.	**Quelle contrainte de conception faut-il respecter ?** Appliquer moindre privilège, validation, frontières explicites et échec sûr.
How should the behavior be verified? Test malicious input, unauthorized access and denial-of-service limits.	**Comment vérifier le comportement ?** Tester entrée hostile, accès interdit et limites de ressources.

ENGINEERING NOTE / NOTE TECHNIQUE

ENGINEERING NOTE / NOTE TECHNIQUE: Reliability and security interact: retry limits prevent resource exhaustion, integrity protects recovery state, and fail-safe design handles unknown commands. / Fiabilité et sécurité se rejoignent: retries bornés, état récupéré intègre et refus sûr des commandes inconnues.

CODE / API EXAMPLE

```
policy={"max_retries":3,"require_checksum":True,"safe_default":False}
assert policy["max_retries"]<=3 and policy["require_checksum"]
```



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